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REVIEW ARTICLE

Advances in sustainable technologies for industry and environment for a greener future

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Abstract

Accelerating climate change and resource degradation demand technological pathways that simultaneously reduce emissions, enhance efficiency, and build resilience. This review systematically synthesizes recent peer-reviewed advances in sustainable and green technologies across five domains: energy, materials and construction, water and waste management, manufacturing, and nature-based urban systems. Literature published between 2014 and early 2025 was retrieved from Scopus, Web of Science, IEEE Xplore, and Elsevier databases using domain-specific search strings covering renewable energy, low-carbon materials, circular economy, smart manufacturing, and urban sustainability. Studies were included if they presented validated models, life-cycle assessments, or technologies with a technology-readiness level ≥ 4. Findings reveal rapid progress in multiple sectors: photovoltaic and hybrid microgrids delivering community-level cost and equity gains (NPV €2042–8195 kW^{-1}), recycled ceramic and foam-glass aggregates reducing embodied carbon without compromising strength, microalgae wastewater systems removing over 90% of nutrients while generating biomass, digital twins and green IoT reducing industrial downtime by ~ 30%, and sponge-city measures increasing floodwater retention by ~ 70%. A cross-sector synthesis identifies enabling mechanisms—policy instruments (PPAs, subsidies, carbon pricing), financial incentives, and digital MRV systems—that lower transaction costs and support large-scale adoption. Technologies are further classified by deployment horizons as fast wins (≤ 2030), no-regrets (to 2040), and long-lead transformations (to 2050). Overall, despite strong technical maturity, diffusion remains constrained by misaligned governance, investment, and circular-resource frameworks. The review proposes an integrative framework linking technology, policy, and finance to accelerate sustainable industrial and urban transitions.

Keywords Sustainable development · Green technologies · Renewable energy integration · Circular economy · Low-carbon materials · Climate resilience · Resource efficiency

Introduction

Sustainable development is a holistic approach that seeks to harmonize economic growth, environmental stewardship, and social inclusion, ensuring that present needs are met without compromising the ability of future generations to meet theirs. Within this framework, green technologies have emerged as critical innovations that reduce environmental harm and enhance efficiency. They minimize pollution, conserve resources, and promote resilience across sectors while driving a paradigm shift toward long-term regenerative solutions. Applications are visible in diverse domains such as sustainable agriculture through GIS-based modelling (Sarkar et al. 2023), clean manufacturing enabled by additive techniques (Wei et al. 2025), low-impact construction materials like recycled cementitious composites (Li et al. 2025), renewable-based energy systems utilizing wind and hydrogen power; Prenzel et al. 2025), and climate-resilient infrastructure exemplified by sponge cities (Soni et al. 2025). Together, these advances reflect a shift from incremental improvements to integrated strategies that support the goals of sustainable development.

Industries and environmental systems are progressively incorporating such technologies to transition toward greener, low-emission operations. In the energy sector, zero-emission

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utility systems for chemical plants integrating thermal energy storage and hydrogen provide pathways to decarbonization while ensuring operational continuity (Prenzel et al. 2025). The construction industry, once considered resource-intensive, is adopting sustainable building materials such as foam glass aggregates derived from recycled glass (Osfouri et al. 2025) and engineered composites using recycled fine aggregates and low-carbon binders (Riaz et al. 2025). Agriculture is undergoing similar transformations, with digital tools like IoT, AI, and augmented reality shaping Agriculture 4.0 to strengthen food security in emerging economies (Alam et al. 2023). These practices, supported by the 5R principles of repair, reuse, refurbish, rebuild, and recycle, along with green dashboards and life cycle assessment tools, demonstrate how circular economy models can enhance both environmental and economic performance (Haleem et al. 2023).

The role of sustainable technologies extends beyond industry to broader social and environmental objectives, aligning directly with global agendas such as the United Nations Sustainable Development Goals (SDGs). Decarbonization frameworks for cement plants have enabled industries to systematically evaluate CO_2 reduction strategies, making significant contributions to net-zero commitments (Khalil and AbouZeid 2025). In the building sector, life cycle assessments of residential HVAC systems provide benchmarks for reducing embodied and operational carbon emissions (Gergely et al. 2025). Circular economy practices utilize recycled ceramic aggregates in cement mortar, reducing waste and conserving natural resources. These aggregates enhance durability by refining microstructure and improving resistance to moisture and chemical attack. Research by Li et al. (2025) confirms that partial ceramic aggregate substitution boosts mortar longevity and strength. This sustainable approach advances both environmental and material performance. Hybrid microgrid models integrating renewable energy with storage systems offer sustainable and cost-effective solutions for remote communities, reducing emissions while improving access to energy and water resources (Valencia-Díaz et al. 2025). Similarly, frameworks for net-positive housing within the Water-Energy-Food-Environment Nexus provide sustainable living solutions for vulnerable populations (Cotella et al. 2025). These cases illustrate how sustainable technologies underpin resource efficiency, equity, and resilience, strengthening the foundations of sustainable development.

The path of these innovations has been shaped by both technological progress and policy support. In the early 2000 s, sustainable technologies were primarily reactive, developed to comply with emission control laws and efficiency standards introduced by governments across Europe, North America, and Asia. Although fragmented, these measures provided the foundation for later systemic approaches. Between 2010 and 2020, sustainability evolved into a more integrated concept that emphasized circular economy practices, renewable energy adoption (Fig. 1), and digitalization. International frameworks such as the Paris Climate Agreement and the SDGs played a central role in directing countries toward low-carbon transitions. During this time, industries embraced strategies based on the 5R principles, supported by government incentives and corporate responsibility initiatives (Haleem et al. 2023). Agricultural innovation also advanced through spatial modelling and digital platforms, particularly in emerging economies that adopted Agriculture 4.0 programs with policy backing (Sarkar et al. 2023; Alam et al. 2023).

The period from 2020 onward has seen the rise of highly integrated, interdisciplinary solutions, often developed through collaboration between governments, industries, and researchers. National strategies have included structured frameworks for decarbonizing cement plants (Khalil & AbouZeid 2025), investments in zero-emission utilities powered by renewable energy (Prenzel et al. 2025), and implementation of sponge city concepts for climate-resilient urban planning (Soni et al. 2025). Public initiatives have also supported net-positive housing models in developing regions (Cotella et al. 2025) and hybrid microgrid systems for underserved communities (Valencia-Díaz et al. 2025). These advances highlight how sustainable technologies are being accelerated by regulatory measures, green stimulus packages, and international financing mechanisms.

Government leadership remains central to sustaining this momentum. Policy interventions in the form of emissions trading schemes, tax reliefs, subsidies, and public–private partnerships have proven effective in driving innovation and adoption. At the same time, future developments are expected to be shaped by global reporting standards, digitization of environmental governance, and cross-sectoral collaboration. The historical progression of sustainable technologies thus reflects a transformation from isolated compliance-based approaches to systemic, policy-driven, and innovation-led frameworks that integrate economic, environmental, and social priorities.

Sustainable and green technologies are increasingly recognized as essential for mitigating climate change, conserving resources, and maintaining long-term ecological balance. Beyond their technical functions, these innovations act as catalysts for systemic transformation across industries, communities, and governance structures (Fig. 2). In this review, *nature-based solutions* (*NbS*) refer to ecosystem-based interventions—such as sponge-city design and microalgae-driven wastewater treatment—that simultaneously address environmental and societal challenges. *Circular flows* describe material and resource pathways that extend product lifecycles through reuse, repair, remanufacture, and recycling, while *digital monitoring tools* encompass

Fig. 1 Renewable energy sustainability

IoT sensor networks, digital twins for system optimization, and MRV platforms for standardized performance tracking, and other related domains were studied. To capture recent advancements in these domains, the review systematically analyzed the latest peer-reviewed studies retrieved from Scopus, Web of Science, and ScienceDirect using targeted keywords related to sustainability, green technologies, resource efficiency, and ecological impact. Studies were included if they presented validated models, life-cycle assessments, or technologies with a technology-readiness level ≥ 4. Key technical parameters were extracted from each study to examine practical applications and emerging innovations. This structured synthesis reveals how sustainable technologies enhance efficiency, reduce ecological footprints, and strengthen resilience. By aligning technological progress with policy priorities and societal needs, they collectively advance global sustainability goals and support the transition toward a more inclusive, low-carbon future (Fig. 3).

Literature reviewed

Survey of existing research

This section reviews prior studies across key sustainability domains, consolidating findings and methodological insights to map current knowledge. The study aims to develop deep insight into these domains by identifying gaps and guiding targeted future research.

Renewable energy

Sharma et al. (2011) have reviewed the techno-economic analysis and modelling of stand-alone versus grid-connected small hydro-power systems, aiming to evaluate their role in meeting rural and remote energy demands cost-effectively while addressing challenges of reliability, efficiency, and sustainability. The study synthesizes a wide

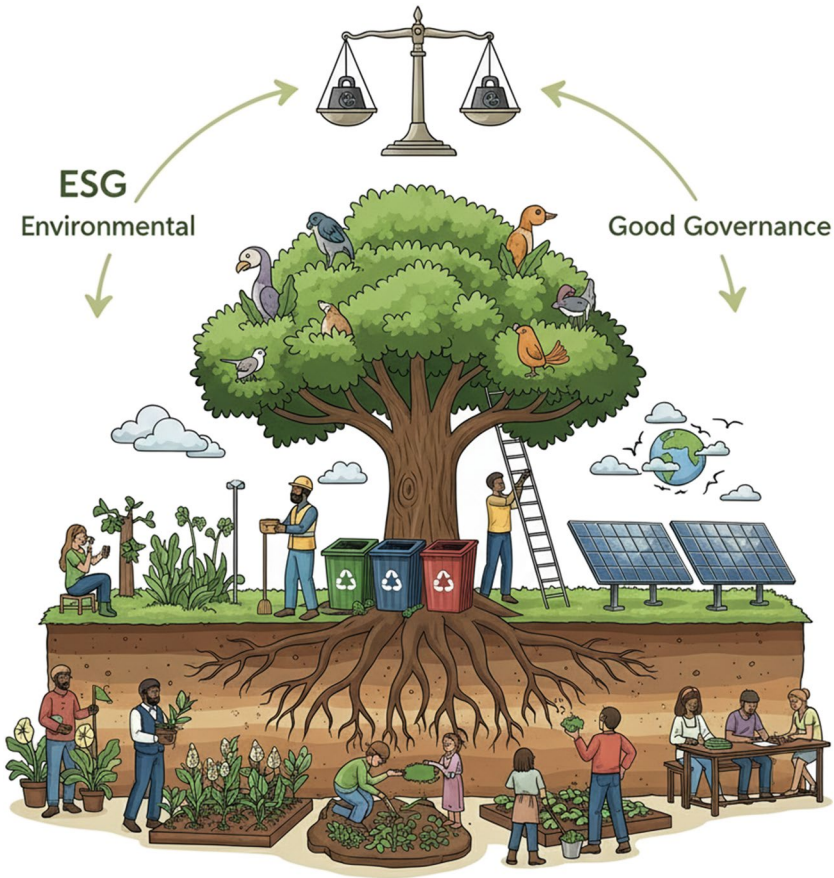

Fig. 2 ESG sustainability

range of research on small hydro systems, highlighting materials such as induction and synchronous generators, electronic load governors, and hybrid renewable configurations. Equipment models include self-excited induction generators, parallel induction generator schemes, and microprocessor-based governors. Process parameters discussed range from electric power delivery factor (EPDF) values between 0.0 and 1.0, backup ratios approaching 100% for stand-alone schemes, and transmission line losses that limit grid extension feasibility in remote terrains. The reviewed results indicate that stand-alone systems offer reliability and decentralized electrification for isolated areas but require costly backup and storage, while grid-connected systems reduce fossil fuel dependency and improve overall efficiency but face challenges of integration and line losses. A major technical insight is that modelling and simulation tools can optimize system cost by balancing demand, resource availability, and equipment sizing, while hybrid solutions enhance resilience. The synthesis emphasizes that small hydro, if integrated through innovative control, simulation, and cost–benefit analysis, can substantially contribute to rural electrification, sustainable development, and climate change mitigation.

Kumar and Majid (2019) have reviewed renewable energy for sustainable development in India: current status, future prospects, challenges, employment, and investment opportunities, aiming to assess how renewable adoption supports energy security, climate mitigation, and economic growth. The study examines solar, wind, biomass, hydropower, waste-to-energy, and nuclear as complementary options for India's growing energy demand. Materials and technologies covered include solar PV systems, wind turbines, bioenergy conversion units, and small hydropower schemes. Equipment discussed includes grid integration systems, renewable purchase obligation frameworks, and hybrid setups for rural electrification. Process parameters presented include India's renewable targets of 175 GW by 2022 (100 GW solar, 60 GW wind, 10 GW bio, 5 GW hydro), projections of 256 Mtoe renewable energy consumption by 2040 (up from 17 Mtoe in 2016), and renewable shares rising from 2% in 2016 to a predicted 13% by 2040. Results highlight strong government policy support, private sector leadership (95% of renewable installations), and significant job creation potential—330,000 new jobs by 2022. Key technical insights include India's estimated renewable potential of ~ 1096 GW (solar 748 GW, wind 302 GW, small hydro 19.7 GW, biomass 17.5 GW) and the crucial role of regulatory frameworks in overcoming intermittency, financial, and infrastructural barriers. The study emphasizes that coherent policy measures, private investment, and R&D-driven

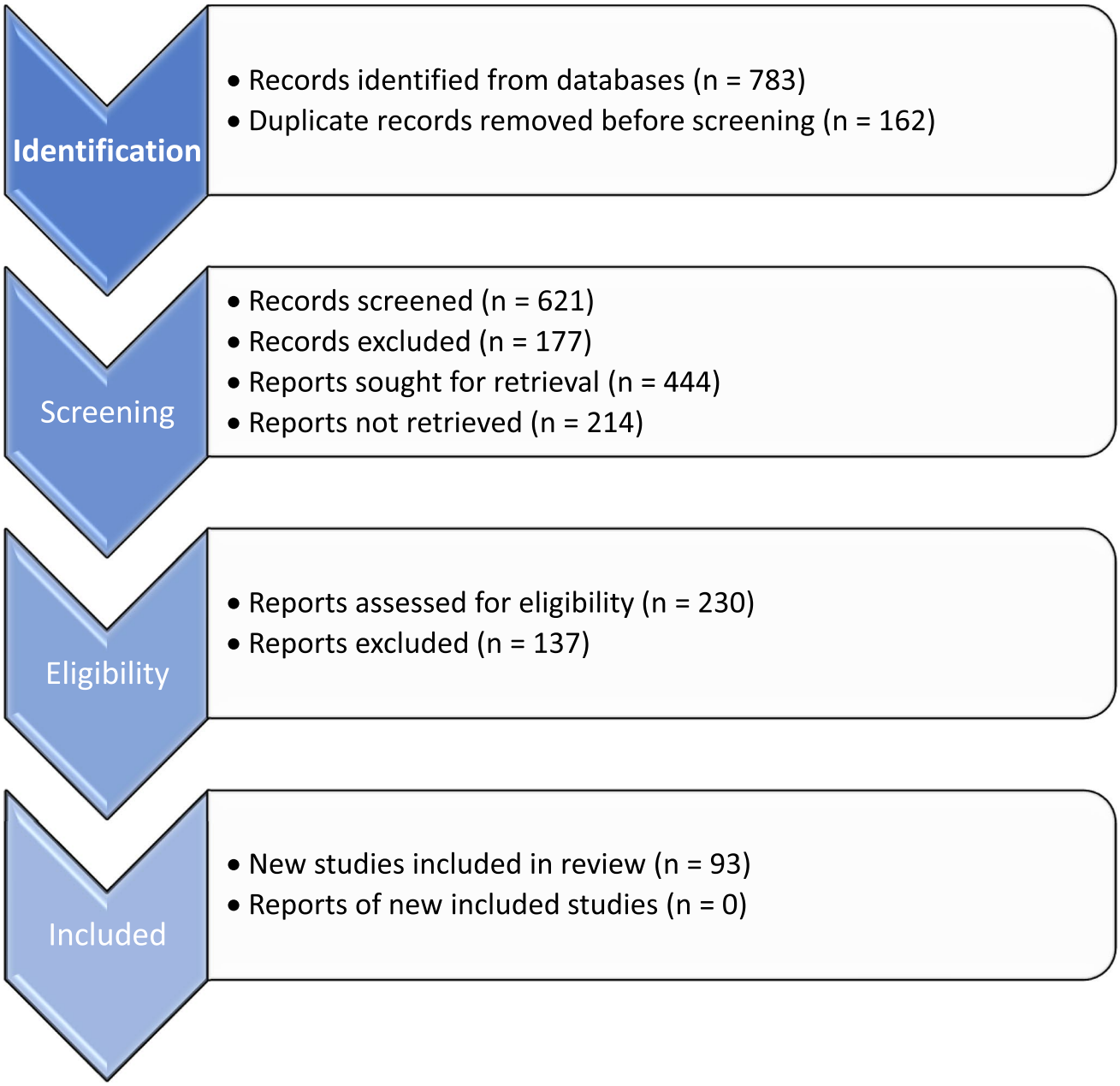

Fig. 3 Prisma flow diagram

efficiency improvements can enable India to become a global leader in clean energy.

Ang et al. (2022) have presented a comprehensive study of renewable energy sources: classifications, challenges, and suggestions, aiming to review the current status of renewable energy technologies and their potential to replace fossil fuels while identifying limitations and proposing solutions. The study discusses six major renewable sources—solar, hydro, wind, bioenergy, geothermal, and hydrogen—along with hybrid renewable systems. Materials covered include photovoltaic (PV) cells, concentrated photovoltaic systems, hot carrier converters, multi-junction solar cells, floating PV, porous structure receivers, and nanofluids in solar thermal collectors, as well as wind turbines, bio-digesters, geothermal plants, and hydrogen electrolyzers. Equipment highlighted ranges from solar concentrators and PV/T modules to offshore and onshore wind turbine systems. Process parameters reported include solar cell efficiencies up to 40.7% for multi-junction systems, hot carrier converters reaching ~ 65%, concentrated PV from 1 to 2000x, solar thermal absorption above 90%, and levelized cost of energy (LCOE) reductions up to 85% for concentrating solar power between

2010 and 2020. Test results show that while renewable technologies have improved efficiency and reduced costs significantly, intermittency, geographic dependency, and high initial costs remain barriers. A major technical contribution lies in cataloguing hybridization strategies and advanced materials that can enhance energy conversion efficiency and system reliability. The study concludes that integrating renewables with storage, cost-reduction policies, and hybrid technologies can accelerate the transition to sustainable, clean, and economically viable electricity generation.

Anwari et al. (2022) have investigated the impact of renewable energy integration on Malaysia's electricity market and carbon emissions, aiming to analyze how increasing renewable energy penetration affects generation costs, carbon output, and market dynamics. The study uses Malaysia's Single Buyer Market (SBM) as a case setting, employing data on generation capacity, fuel mix, demand profiles, and renewable energy targets. Materials considered include solar PV, biomass, small hydro, and gas- and coal-fired plants, with equipment references limited to power system modeling tools. Process parameters include renewable penetration levels ranging from current baselines to over 30%, generation cost per MWh, and CO_2 emissions in metric tons. Simulation results show that increasing renewable share reduces reliance on coal, lowers marginal generation costs, and significantly cuts carbon emissions—up to 30% reduction at higher penetration levels. The major technical insight is that system flexibility and grid stability become increasingly important with higher variable renewable inputs, requiring investment in storage, grid upgrades, and flexible generation. The analysis demonstrates that targeted renewable expansion, combined with supportive policy and market mechanisms, can transform Malaysia's electricity sector into a low-carbon, cost-competitive system while meeting sustainability commitments.

Al Mufti et al. (2025) have investigated the experimental validation and parametric optimization of a thermoacoustic refrigerator (TAR) as a greener cooling solution leveraging acoustic energy, aiming to enhance its performance through configuration refinement and thermal management. The study focused on a standing wave TAR setup composed of a resonance tube, heat exchangers, a loudspeaker-based acoustic driver, and a stack made of low thermal conductivity materials with high heat capacity. Helium was used as the working fluid for its favorable thermophysical properties. Equipment included a full-scale laboratory experimental rig for measuring pressure amplitudes and temperature gradients, coupled with high-fidelity numerical simulations combining global and local domains. Process parameters explored involved frequency (variable), stack position (normalized from 0.175 to 0.33), stack spacing (3.3 δk), blockage ratio (0.7), normalized stack length (0.24), and drive ratio (ranging from 1.5 to 6.4%). Experimental results recorded a maximum temperature difference of 67 °C at a 6.4% drive ratio, while simulations predicted up to a 124 °C temperature drop. Performance analysis revealed that higher drive ratios decreased the coefficient of performance (COP) at lower temperature gradients but increased cooling power. The major technical highlight is the correlation between normalized design parameters and acoustic-thermal coupling, with precise stack tuning critical to achieving high cooling efficiency. The study confirms TARs' potential as near-zero global warming technologies for sustainable cooling, supporting applications from room temperature to cryogenics without harmful refrigerants or significant emissions.

Basilico et al. (2025) have analyzed the economic performance and equitable benefit distribution of photovoltaic-based renewable energy communities (RECs) for sustainable cities. The study focuses on optimizing subsidies, market participation, and profit-sharing models to encourage prosumer engagement and reduce energy inequality. The methodology employed discounted cash flow analysis using net present value (NPV) as the key metric under two market conditions—low and high—alongside simulations of residential REC setups with four renewable self-consumers (RSCs). Photovoltaic (PV) systems were used as the primary energy source, with system performance influenced by investment costs, self-consumption ratios, and energy price dynamics. Results showed strong financial viability, with NPV per kW ranging from €2042 to 5148 (low market) and €3347 to 8195 (high market), and profit distribution varying between 15 and 35% among RSCs. Two innovative profit distribution models were introduced: one incentivizing energy-conscious behavior through lower internal trading prices, and another focused on income-based equity by allocating more benefits to low-income members. Both models demonstrated a reduction in cost disparity and encouraged participation regardless of energy consumption levels. Inclusion of non-producer consumers in RECs further diversified benefit allocation, enhancing social inclusivity and reducing load on the public grid. By aligning environmental resilience with social justice, the proposed models offer scalable blueprints for equitable urban energy transitions, reinforcing the role of RECs in achieving SDGs 7 and 11.

The integration of insights from above diverse research suggests that advancing sustainable energy systems requires an integrated approach where technical optimization, system modelling, and socio-economic mechanisms operate in concert to deliver both efficiency and equity. In small hydropower, techno-economic modelling demonstrates that factors such as the electric power delivery factor, ranging from 0.0 to 1.0, high backup ratios, and transmission losses define whether standalone, grid-connected, or hybrid systems are viable, with hybrids offering greater resilience. Expansions in India, with an estimated 1096 GW renewable potential, and Malaysia, where simulations show renewable

penetration beyond 30% can cut CO_2 emissions by up to 30%, illustrate how large-scale adoption hinges on grid integration, storage capacity, and strong policy frameworks. At the technology frontier, thermoacoustic refrigeration proves a viable green cooling method where careful tuning of drive ratio between 1.5 and 6.4%, stack position from 0.175 to 0.33, and spacing at 3.3 δk yields temperature differences up to 124 °C, underscoring the significance of acoustic thermal coupling in achieving efficiency without harmful refrigerants. On the socio-technical side, photovoltaic-based renewable energy communities demonstrate economic viability with net present values between €2042 and €8195 per kW, while equity-oriented profit-sharing models reduce disparities and expand inclusivity, making energy transition more socially just. Complementary advances show that solar cell parameter extraction techniques, from simple one-diode to hybrid models, enable more accurate determination of Rs, Rsh, n, and Iph, critical to reliable PV system modelling, while wind energy harvester assessments highlight the role of multidimensional modelling in predicting load-driven structural risks under variable conditions. Taken together, these contributions establish that building resilient low-carbon energy futures requires synchronizing parameter-level precision, high-fidelity system modelling, and equitable market frameworks so that technological advances translate into scalable, sustainable, and inclusive energy transitions.

Hydrogen sustainability

Sadik-Zada et al. (2023) have investigated the sustainable management of lithium and green hydrogen and long-run perspectives of electromobility by analysing lithium's strategic role in the decarbonization of the transport sector. The study evaluates the challenges and limitations of lithium-ion battery (LIB)-based electromobility using existing datasets, with a focus on resource constraints, recycling capabilities, and environmental degradation. Materials used include lithium and hydrogen, while no specific experimental equipment is used, as the research is conceptual and analytical in nature. The research discusses lithium's properties, highlighting its electrochemical potential (3.05 V) and specific capacity (3860 Ah/kg), as well as its global production, demand trends, and extraction impacts. Process parameters discussed include lithium availability, recycling potential, and market dynamics. Results demonstrate that LIBs, while currently pivotal for battery electric vehicles, face sustainability challenges due to resource scarcity, geopolitical supply risks, and weak recycling frameworks. The paper juxtaposes LIBs with hydrogen fuel cells, indicating the latter as a more sustainable long-term solution that aligns better with circular economy principles and strong sustainability goals. Hydrogen fuel cell electric vehicles (FCEVs) emerge as superior due to their minimal environmental footprint and greater alignment with long-term decarbonization and circular economy models, thus suggesting a gradual transition from LIB to hydrogen-based mobility.

Chitgar and Sadrzadeh (2025) have investigated the integration of geothermal energy with desalination and hydrogen production systems for sustainable energy and water supply. The purpose of the study is to compare the performance of two configurations—Multi-Effect Distillation (MED) and Reverse Osmosis (RO)—when powered by geothermal energy, for both desalinated water and hydrogen production. The system materials include geothermal heat as the primary energy source, desalinated water as feedstock, and electrolyzers (Proton Exchange Membrane Electrolyzers) for hydrogen generation. Equipment used includes MED units, RO systems, and hybrid desalination setups, all evaluated under varying operational parameters such as geothermal heat flow, system pressures, and energy conversion efficiencies. Through artificial neural networks (ANN) and genetic algorithm (GA)-based multi-objective optimization, the study assesses net power output, hydrogen yield, water recovery rates, exergy efficiency, and levelized costs of hydrogen (LCOH) and water (LCOW). Results indicate that the RO-based configuration achieves broader optimal conditions and lower costs, while the MED setup offers advantages under specific thermal scenarios. The case study focused on Canada demonstrates the feasibility for supporting both residential water demand and hydrogen-fueled transport, with RO enabling lower costs and higher water production efficiency. A key technical insight is that integrating geothermal energy not only enhances system sustainability but also extends PEM electrolyzer life by using high-quality desalinated water, reinforcing the system's viability for circular energy-water applications.

Fuhrlaender et al. (2025) have proposed an integrated operating concept for the green hydrogen transformation of the iron and steel production system. The research focuses on transforming the conventional carbon-intensive blast furnace-basic oxygen furnace (BF-BOF) route into a low-emission direct reduction-electric arc furnace (DR-EAF) process powered by green hydrogen produced via water electrolysis (WE) and renewable energy (RE). The study incorporated an extended production system comprising WE units, renewable energy sources (notably offshore wind via HVDC link), electric arc furnaces (EAF), combined cycle gas turbines (CCGT), and a connected consumer, evaluated under three scenarios using Python-based simulations. The equipment and process parameters included large-scale electrolyzer operations governed by load prioritization strategies, with constraints based on gradient and power limits, emphasizing internal system balance and power grid stability. Under Scenario 1, prioritizing hydrogen production, maximum WE operation was achieved, while Scenario 2 focused on minimizing grid load by adjusting the WE to only operate under

negative residual load conditions. Results demonstrated the system's ability to reduce Scope I and II CO_2 emissions significantly, optimize internal energy use, and relieve stress on the upstream power system. The integrated operating concept (IOC) allowed flexible system responses, notably fast down-ramping of the electrolyzer during grid peaks and prioritization of critical loads, creating a prosuming entity with centralized control. The approach highlights the feasibility of using hydrogen flexibility to achieve environmental goals, stabilize electricity demand, and transition industrial systems into sustainable, economically attractive, and self-balancing configurations.

Hopken et al. (2025) have modelled green hydrogen storage in salt caverns to analyze the operational implications of future storage demands based on renewable energy scenarios, with a focus on Germany's evolving power mix and infrastructure requirements. The study aimed to understand how different hydrogen demand profiles—driven by wind-heavy (Lower Saxony) and PV-heavy (national) electricity compositions—affect storage system design and efficiency. Materials modeled included renewable electricity generation datasets for the year 2045, cavern specifications from STORAG ETZEL GmbH, and thermodynamic gas properties derived using CoolProp. The custom-built simulation model incorporated parameters such as mass flow rate (kg/s), cavern volume, pressure, heat flux, and temperature gradients. No physical experiments were conducted; rather, the simulation platform solved ordinary differential equations to iteratively calculate cavern states across time steps. Results indicated that the wind-dominant scenario exhibited high turnover rates, necessitating larger compressors and resulting in steeper pressure and temperature gradients within the cavern, while the PV-dominant case showed smoother operation but required more adaptable surface components. The major technical achievement lies in the development of a flexible and thermodynamically robust model capable of predicting the performance of hydrogen caverns under future power conditions. This model supports infrastructure planning by identifying the operational adaptations needed to meet future hydrogen storage demand with optimized cavern design and minimal energy losses.

Huang et al. (2025) have assessed the sustainability of hydrogen production via polymer electrolyte membrane water electrolysis (PEMWE) using various configurations of solar photovoltaic-battery-grid systems in China, aiming to determine the optimal system in terms of environmental and economic performance. The configurations included standalone photovoltaic (SPV), off-grid photovoltaic with battery storage (BPV), and grid-connected photovoltaic (GPV), each supplying power to PEMWE units. Materials involved consisted of solar PV modules, lithium-ion batteries, and PEM electrolyzers, while the life cycle assessment (LCA) and life cycle costing (LCC) were used as analytical tools. Equipment and simulation tools employed included impact modeling software based on the ReCiPe and CML methods, with key process parameters covering climate change impact (4.3–6.5 t CO_2 eq./t H_2 for SPV), levelized cost of hydrogen (3953–6029 $/t for SPV, 2741–4048 $/t for GPV, and over 11,000 $/t for BPV), and multiple environmental impact categories such as land use, ecotoxicity, and metal depletion. Test results showed SPV had the best environmental profile, outperforming coal gasification in nine categories and steam methane reforming in four, though it was costlier than GPV. GPV offered the lowest cost but a higher environmental burden. The major technical finding is that system configuration strongly influences sustainability trade-offs, and that under current and projected energy scenarios, SPV is most environmentally favorable, while GPV remains more economically viable, guiding future regional hydrogen infrastructure planning in China.

Jin et al. (2025) have developed a GIS-based framework for the design of hybrid energy storage systems (HESS) for steady green hydrogen production in residential areas, addressing challenges related to the intermittency of renewable energy sources and degradation of electrolyzers due to fluctuating operations. The purpose of this study was to create an optimized, location-specific solution that ensures constant-rated operation of alkaline electrolyzers powered by photovoltaic (PV) panels and supported by lithium-ion batteries. The framework integrates a mixed-integer linear programming (MILP) optimization model with spatial data from GIS analysis to match energy demand profiles of Italian residential users with typical loads ranging from 1.5 to 3.0 kW. Materials used include 3.0–4.5 kW PV systems, lithium-ion batteries with capacities of 6.0–7.0 kWh, and electrolyzers sized at 100–260 W. Process parameters involved maintaining a 95–100% electrolyzer load factor, ensuring steady hydrogen production while minimizing battery cycling and degradation. The system was tested for daily electrical loads of 2.8–6.0 kWh and evaluated economically with hydrogen cost recovery between €12 and 19/kg, while offering CO_2 emission reductions of 0.35–0.83 tons per user annually. Results emphasized the dominant cost contribution of batteries (60% of system CAPEX) and the importance of maintaining constant operating conditions to avoid performance decay in electrolyzers. By integrating technical feasibility, economic viability, and environmental sustainability, the model provides a replicable strategy for deploying decentralized, off-grid hydrogen systems in residential contexts aligned with carbon reduction targets.

Roy et al. (2025) have investigated the optimization of hybrid energy systems (HESs) for cost-effective green hydrogen production in remote Australian communities. The study aimed to assess the techno-economic performance of two system configurations: Type-A (PV/Batt/FC/EL/HT) and Type-B (PV/FC/EL/HT), using various metaheuristic

algorithms such as NSGA-II, SQPA, CSA, FPA, CPSO, and HSA to achieve optimal energy reliability and affordability. The primary materials and components include photovoltaic (PV) panels, batteries, fuel cells (FC), electrolyzers (EL), and hydrogen tanks (HT), with a key emphasis on optimizing the PV tilt angle. Among these, the NSGA-II optimized Type-A system, integrating a battery, achieved the best economic results with a net present cost (NPC) of $226,500, levelized cost of electricity (LCOE) at $0.193/kWh, and levelized cost of hydrogen (LCOH) at $4.88/kg. System performance was found to be highly sensitive to the tilt angle of PV panels, where the optimal angle of 30° significantly reduced costs; increasing it to 70° raised system costs by up to 75%. Moreover, enhancing the efficiency of fuel cells from 40 to 60% led to reductions in NPC, LCOE, and LCOH by $40,000, $0.04/kWh, and $0.1/kg, respectively. Hydrogen autonomy was favored in Type-B due to higher FC (18.63 kW) and EL (14 kW) capacities, enabling a maximum hydrogen yield of 4737 kg/year under SQPA optimization. The study demonstrates how optimized system design, especially with appropriate PV tilt and component efficiency adjustments, can reduce costs by up to 40%, offering a resilient model for integrating renewable electricity and hydrogen production in off-grid, remote settings.

Overall, the research on hydrogen sustainability highlights its growing role as a versatile, low-carbon energy carrier across multiple sectors. Hydrogen and fuel cells are increasingly applied in transport, industry, heat, and electricity storage, offering advantages over lithium-ion battery-based solutions, particularly for heavy-duty mobility and circular economy alignment. Key challenges include scaling infrastructure, optimizing production costs, and ensuring a steady supply from renewable sources. Studies demonstrate diverse production pathways, including water electrolysis powered by solar, wind, or geothermal energy, often integrated with energy storage systems to manage intermittency and maintain consistent electrolyzer operation. Hybrid systems combining photovoltaic panels, batteries, and fuel cells can achieve cost-effective, reliable hydrogen supply in residential and remote contexts, while system configuration, component efficiency, and site-specific factors strongly influence economic and environmental performance. Industrial applications, such as green hydrogen in steel production, show potential for deep decarbonization by replacing carbon-intensive processes and enabling flexible grid interactions. Large-scale storage solutions, such as salt caverns, support renewable integration and balance variable demand. Overall, hydrogen emerges as a critical component for sustainable energy transitions, complementing electrification, enhancing system flexibility, and enabling long-term decarbonization across transport, industry, and energy systems.

Carbon neutrality

Chen (2021) have discussed carbon neutrality as a global imperative for sustainable development, aiming to analyse its necessity, challenges, and opportunities by placing it within the historical context of industrial revolutions and energy transitions. The study highlights fossil fuels as the main material contributors to greenhouse gas accumulation and emphasizes the role of renewable energies such as wind, solar, biomass, tidal, and nuclear, along with carbon sinks through afforestation and soil management. Equipment and mechanisms mentioned include carbon capture and storage (CCS) systems, carbon markets, and renewable energy infrastructures. Process parameters involve achieving carbon neutrality by 2050–2060 to limit global temperature rise to 1.5–2 °C, carbon pricing at ~ US$52/t$CO_2$ to drive innovation, and renewable adoption sufficient to replace fossil resources within decades. Test results, including historical climate records, show a 1.2 °C rise in global average temperature since 1850 and sea level increases of 0.24 m, with projections of 0.3–1.5 m by 2100 under different emission scenarios. A critical technical insight is that carbon neutrality represents not only an environmental milestone but also a potential "fifth industrial revolution," reshaping energy systems, economies, and societies toward long-term sustainability.

Zhao (2022) has introduced carbon neutrality as a multidisciplinary research and innovation frontier, emphasizing the creation of a dedicated journal platform to integrate diverse perspectives across energy, environment, finance, and policy. The paper highlights low-carbon technologies such as renewables, energy storage, and CCUS, while also addressing complementary areas like carbon finance, circular economy, and carbon management. Materials and systems include renewable energy plants, financial mechanisms for green investment, and ecosystem governance frameworks. Process parameters discussed involve global commitments under the Paris Agreement to achieve carbon neutrality in the second half of the century and research efforts aimed at bridging gaps between technical and social disciplines. Results underline that while numerous journals cover single aspects of low-carbon research, the absence of a comprehensive, cross-disciplinary platform has hindered rapid innovation and dissemination. The technical significance of this work lies in launching the Carbon Neutrality journal in 2021, which serves as a global hub for advancing scientific, technological, and policy knowledge exchange to accelerate progress toward net-zero goals.

Pan and Sun (2023) have analyzed the nature and rationale of carbon neutrality from perspectives of environmental sustainability, economics, and energy security, aiming to clarify conceptual misunderstandings and guide China's long-term transition toward a net-zero future. Materials

emphasized include fossil fuels as short-term stabilizers, disruptive zero-carbon energy technologies (wind, solar, nuclear, hydrogen), and carbon capture and carbon sinks as supplementary measures. Equipment and systems include large-scale renewable installations, energy-efficient infrastructure, and financing frameworks for zero-carbon investment. Process parameters outlined include phasing out fossil fuel dominance from ~ 84% of China's energy mix in 2021 to below 20% by 2060, raising non-fossil energy shares to 25% by 2030 and over 80% by 2060, and implementing stepwise policies rather than abrupt shifts. Results show that fossil fuel dependence is the main obstacle to neutrality, with CO_2 emissions from coal, oil, and gas accounting for over 10 Gt annually in China. A major technical aspect lies in identifying fossil fuel–derived carbon as climate-catastrophic and demonstrating that only disruptive zero-carbon energy and structural reforms can eliminate this dependency while ensuring energy security and economic resilience. The analysis concludes that carbon neutrality is not only an environmental necessity but also a strategic opportunity for sustainable economic growth and global competitiveness.

Shah et al. (2023) have developed hydraulic modelling of catalytic converter geometry for automotive exhaust pollutant reduction, focusing on optimizing flow uniformity and minimizing pressure drop to enhance converter performance and lifespan. The study used ANSYS Fluent for 2D axisymmetric CFD simulations across 16 different geometries of catalytic converters composed of diffuser, inlet pipe, monolith substrate, and outlet pipe. No experimental equipment was deployed; simulations incorporated finite volume methods with varying diffuser angles and cone shapes. Key process parameters included velocity profiles, pressure drop, uniformity index, and flow expansion characteristics, with numerical results showing that optimized diffuser geometry significantly reduces reverse flow and eddy formation. Improved designs led to better flow distribution across the monolith, contributing to delayed ageing and enhanced conversion efficiency for pollutants like CO, NOx, and HC. The study demonstrated how geometric optimization directly correlates with emission control, supporting cleaner combustion in conventional vehicles amidst increasing environmental regulations.

Wang et al. (2024) have explored carbon neutrality pathways in China with a focus on energy transition, technological innovation, and policy design, aiming to provide a roadmap for achieving China's dual goals of carbon peaking before 2030 and neutrality before 2060. The study reviews materials such as renewable energy technologies—solar, wind, hydro, biomass, and nuclear—alongside carbon capture, utilization, and storage (CCUS) systems, and hydrogen as an energy carrier. Equipment and systems discussed include renewable power plants, smart grids, carbon trading platforms, and CCUS infrastructure. Process parameters emphasized include scaling renewables to meet over 80% of primary energy demand by 2060, increasing energy efficiency across industries, and implementing high carbon pricing to incentivize clean energy deployment. Test results from modelling and scenario analysis indicate that renewable energy could meet the majority of energy demand if accompanied by rapid technological progress, while coal dependence must drop from over 70% in 2020 to less than 20% by 2060. A major technical point highlighted is the integration of disruptive zero-carbon technologies and industrial upgrading to overcome carbon lock-in and ensure a stable energy supply. The study stresses that coordinated policy, market reforms, and international cooperation are essential for China's successful carbon–neutral transition.

Gad El Mola (2025) has developed a fuzzy-QFD-based framework for integrating green supply chain (GSC) principles into energy production systems, addressing the need for sustainability across production, distribution, and consumption stages. The study aimed to identify and prioritize green requirements (GRs) and green factors (GFs) specific to the energy sector using fuzzy logic and quality function deployment (QFD). Materials involved conceptual components like energy efficiency policies, emission controls, and green process management. Equipment and methodology incorporated the Analytic Hierarchy Process (AHP) via Expert Choice Software for calculating GR weightings, followed by fuzzy-QFD techniques for establishing relationships between GRs and GFs. Process parameters included cost, environmental quality, and operational efficiency, evaluated using technical importance ratings and normalized analysis metrics. Sensitivity analysis validated the model by demonstrating that adoption of green chain management increased environmental performance by 25%, efficiency by 20%, and operational costs by 20%. The framework was experimentally validated through a case study in an energy-producing firm, confirming its adaptability and robustness. The proposed model advances traditional supply chain management by embedding green priorities directly into energy operations and offers a structured path to reduce carbon footprints, meet regulatory standards, and enhance long-term sustainability across energy-intensive sectors.

Khalil and AbouZeid (2025) have investigated a structured assessment framework for decarbonization in cement plants, aiming to bridge the gap between plant-level initiatives and global net-zero emissions (NZE) targets. The study proposes an Excel-based assessment tool built on the 5Cs framework (clinker, cement, concrete, construction (re) carbonation, and carbon capture/storage-utilization) to identify, score, and prioritize CO_2 reduction measures. Materials considered include raw materials and fossil/alternative fuels; equipment and systems span kilns, coolers, CCS units, and energy monitoring tools. Process parameters analyzed include CO_2 emissions per ton of cement (622 kg to 580 kg),

fuel substitution levels, and energy efficiency improvements (up to 18%). The assessment tool incorporates both qualitative expert surveys and quantitative data inputs to compute an overall sustainability score, colour-coded to represent implementation levels. Applied in an Egyptian cement plant case study, the model yielded a sustainability score of 3.501, reflecting strong decarbonization performance. The tool facilitates efficient decision-making, strategic planning, and resource allocation by ranking measures by effectiveness. It enables continuous monitoring, adaptation to regional contexts, and alignment with NZE 2050 goals, proving to be a scalable and structured approach for driving low-carbon transformation in cement manufacturing.

Prenzel et al. (2025) have modeled a zero-emission utility system for chemical sites by integrating power purchase agreements (PPAs) with thermal energy storage (TES) to optimize decarbonization strategies. The purpose was to identify a cost-effective configuration that ensures an uninterrupted energy supply while eliminating greenhouse gas emissions. Materials involved include molten salt (used as TES medium) and green hydrogen as a clean fuel source. The equipment modeled consisted of hydrogen-fired gas turbines, electric heaters, molten salt steam generators (MSSG), two-tank TES systems, and high-pressure steam turbines, all simulated through the TOP-Energy® optimization platform. Key process parameters explored were steam pressure levels (HP, 110 bar at 530 °C; MP, 31 bar at 370 °C; LP, 6 bar at 210 °C), wind and solar power profiles, TES charge/discharge cycles, and hydrogen price variability. Sensitivity analysis showed that TES can reduce annual operational expenditures by up to 27%, even under high hydrogen price scenarios. Integration of TES allowed efficient use of curtailed renewable electricity and reduced dependence on high-cost hydrogen during low generation periods. The model also demonstrated that TES, with a high storage density (~ 1300 kWh_th/m2), is more spatially efficient than battery systems, making it ideal for space-constrained chemical sites. The study concludes that combining PPAs with molten salt TES not only enhances energy flexibility but also provides financial resilience, ensuring feasible and scalable zero-emission transitions for industrial utility systems.

Sprenger et al. (2025) have developed an anticipatory sustainability assessment framework for early-stage material development, demonstrated through bio-based benzoxazine synthesis, aiming to overcome the limitations of conventional life cycle assessment (LCA) in evaluating low-TRL materials. The purpose was to create a semi-quantitative and integrative tool named Mat-STAP, built upon LiSET, SURF-UK, and Green Chemistry principles, to guide decision-making in sustainable material selection. Materials analyzed include various bio-based phenolic feedstocks such as allyl phenol and phloretic acid for the synthesis of benzoxazine polymers. Equipment is analytical and software-based, involving sustainability evaluation matrices rather than physical instrumentation. The process parameters considered were chemical feasibility, environmental compatibility, scalability potential, and stakeholder engagement criteria within a multi-criteria decision-making setup. Test results revealed that traditional LCA was insufficient at this stage due to high data uncertainty, while the Mat-STAP tool effectively identified allyl phenol and phloretic acid as the most sustainable feedstocks. The framework proved effective in visualizing process impacts, prioritizing materials under sustainability metrics, and fostering transparent, collaborative early-stage decision-making. By enabling proactive evaluation and bridging gaps in conventional assessments, Mat-STAP establishes a more resilient path for green material innovation in composite applications.

Overall, research highlights diverse strategies for advancing carbon neutrality demands coordinated technical, operational, and systemic innovation spanning automotive, energy, cement, chemical, and material sectors. In automotive applications, catalytic converter optimization through CFD-based modelling demonstrates that diffuser geometry and flow parameters directly influence velocity profiles, pressure drop, and uniformity index, leading to improved pollutant conversion and extended component lifespan under regulatory constraints. Energy transition studies emphasize that renewables, smart grids, carbon capture systems, and hydrogen integration must scale to over 80% of demand by 2060, supported by policy tools like carbon markets and carbon pricing at US$52/t$CO_2$, while fuzzy QFD-based green supply chain frameworks provide a structured means of prioritizing green requirements, improving efficiency by 20% and environmental performance by 25%. Cement plants employing the 5Cs Excel-based assessment tool can systematically rank CO_2 reduction strategies, lowering emissions from 622 to 580 kg per ton of cement through measures such as 18% energy efficiency improvements and high substitution of alternative fuels, enabling alignment with NZE 2050 trajectories. Industrial utility decarbonization benefits from integrating PPAs with molten salt thermal energy storage, where sensitivity analysis shows operating cost reductions up to 27% and storage densities around 1300 kWh_th/m2, proving spatially efficient alternatives to batteries in chemical sites reliant on continuous steam supply. In parallel, early-stage material innovation is advanced by anticipatory sustainability frameworks like Mat-STAP, which extend beyond conventional LCA to assess bio-based benzoxazine feedstocks such as allyl phenol and phloretic acid against chemical feasibility, environmental compatibility, and scalability, thereby guiding resilient green material development. Together, these contributions demonstrate that carbon neutrality is driven by the convergence of precise geometric design, optimized operational frameworks, advanced storage and supply systems, and anticipatory material assessment,

ensuring that industrial decarbonization pathways remain technically robust, economically viable, and environmentally sustainable.

Sustainable construction

Sarhat and Jansze (2014) have investigated the structural behavior of precast beam-to-column dowel connections under cyclic loading, aiming to evaluate their seismic performance and potential for use in moment-resisting frames. The study used full-scale precast reinforced concrete beam–column subassemblies incorporating straight and hooked steel dowels as connection reinforcement, with high-strength grout for embedment. Equipment included hydraulic actuators for applying cyclic lateral loads, load cells, displacement transducers, and data acquisition systems. Process parameters involved varying dowel configurations, embedment depths (typically 8–12 times dowel diameter), and cyclic drift ratios up to failure. Test results showed that hooked dowels provided superior energy dissipation, higher ultimate strength, and delayed stiffness degradation compared to straight dowels, while both types maintained acceptable ductility levels. The major technical insight is that dowel detailing strongly influences joint capacity, crack distribution, and post-peak behavior, with proper grout confinement and dowel anchorage critical to ensuring seismic resilience. The findings demonstrate that with optimized detailing, precast dowel connections can achieve comparable or superior cyclic performance to monolithic reinforced concrete joints, making them a viable option for earthquake-resistant construction.

Bhardwaj and Kumar (2017) have reviewed the use of waste foundry sand (WFS) in concrete, aiming to assess its feasibility as a partial or full replacement for natural river sand in structural grade concrete to address sustainability issues caused by excessive natural sand extraction. The study compiles data from multiple experimental investigations, detailing WFS as a by-product from ferrous and non-ferrous metal casting industries, consisting mainly of high-quality silica sand with binders and additives such as bentonite, sea coal, and chemical resins. Physical properties include particle sizes predominantly between 0.6 and 0.15 mm, specific gravity ranging from 2.39 to 2.79, and fineness modulus between 0.9 and 2.32. Chemical composition varies but is rich in SiO_2 (76–88%) with Al_2O_3, Fe_2O_3, and minor oxides. Equipment used in the reviewed studies includes sieve analyzers, slump cones, compression testing machines, ultrasonic pulse velocity testers, and permeability testing setups. Process parameters in different studies involved WFS replacement levels from 0 to 100% in concrete mixes, curing durations from 7 to 90 days, and tests under varying exposure conditions. Results across reviewed experiments indicate that optimum WFS replacement of 20–30% enhances compressive, tensile, and flexural strengths, abrasion resistance, sulphate and chloride resistance, and reduces permeability, though workability decreases with higher WFS content due to its finer grading and higher water absorption. At higher replacement levels, strength and durability decline, attributed to excess fines and impurities. Microstructural analysis using SEM and XRD showed dense matrix formation at optimum replacement levels. This synthesis highlights that controlled use of WFS in concrete can conserve natural sand resources, improve certain mechanical and durability properties, and promote sustainable waste management in the foundry industry.

Li et al. (2025) have experimented on microstructure and transport properties of cement mortar incorporating recycled fine ceramic aggregates (CA) to evaluate their feasibility as substitutes for natural sand (NS) in mortar production. The purpose was to explore sustainable waste utilization while improving mortar performance. Materials included ordinary Portland cement (OPC, PI 42.5), NS (quartz-based), and household ceramic waste such as decorative vases and flowerpots, crushed into fine aggregates sized 0.08–3 mm. Equipment and techniques employed were mercury intrusion porosimetry (MIP) for pore analysis and scanning electron microscopy (SEM) for microstructural study. Process parameters encompassed varying substitution ratios of CA for NS (0 to 100%) at a fixed water-to-cement ratio of 0.4 and aggregate-to-cement volume ratio of 2.0. Results revealed optimal mechanical performance at 20% CA substitution, with compressive and flexural strengths improving due to enhanced interfacial bonding. At higher substitution levels, strength declined slightly but remained comparable. Durability indices—gas permeability and water absorption—improved with CA content up to 40%, attributed to reduced critical pore size (1.05 μm) and more refined pore structure. The study established linear correlations between durability and pore parameters, indicating that CA not only narrows pore size distribution but also enhances impermeability. Through the effective use of ceramic waste from Jingdezhen's industry, this approach contributes to ecological conservation and material circularity while producing mechanically and chemically durable mortar composites.

Osfouri et al. (2025) have investigated a sustainable structural lightweight concrete incorporating foam glass aggregates, aiming to substitute natural coarse aggregates with eco-friendly alternatives to reduce environmental impacts while maintaining structural integrity. The study utilized virgin and recycled window glass (particle size < 90 μm) to synthesize foam glass aggregates using a two-stage sintering method with silicon carbide (SiC < 1 μm) as a foaming agent and alumina (Al_2O_3) as a structural modifier. Foam glass aggregates were used in 20%, 40%, and 60% volume replacements of natural coarse aggregates in concrete mixes. Equipment included a QUANTACHROME particle size

analyzer, Rigaku XRD, Carl Zeiss SEM/EDS, and a MicrO-Vis heating microscope for microstructural characterization, along with a universal testing machine (Instron 5566) for compressive strength. Process parameters examined were compressive strength (17 MPa +), bulk density (< 2000 kg/m^3), and alkali-silica reactivity (ASR) assessed using thermogravimetric analysis (TGA). Results demonstrated that a 20% substitution achieved structural-grade strength with significant weight reduction and thermal insulation properties. ASR and TGA confirmed the non-reactivity and thermal stability of foam glass, mitigating long-term durability risks. The findings validate that glass-based lightweight aggregates can be effectively engineered for structural applications, offering a circular economy solution by diverting waste glass from landfills and contributing to eco-efficient concrete production.

Riaz et al. (2025) have developed a novel engineered cementitious composite (ECC) incorporating limestone calcined clay cement (LC3) and recycled fine aggregates (RFA) as a sustainable and ductile construction material. The aim of the study was to overcome environmental and performance limitations of conventional ECC by replacing ordinary Portland cement (OPC) and silica sand with LC3 and RFA derived from waste concrete. Four replacement levels of RFA (0%, 30%, 60%, and 100%) were evaluated to study mechanical performance and material behavior. The materials used included LC3 as a binder and silica sand partially or wholly replaced by RFA. Tests were conducted to assess tensile strength, compressive strength, strain capacity, and microstructural bonding. The results revealed that incorporating 100% RFA improved tensile ductility by 80% and maintained compressive strength with only a 20.5% reduction at 180 days. Interestingly, at 30% RFA, tensile strength increased compared to the control. This performance was attributed to the self-cementing properties of RFA, which reacted with LC3 to form additional hydration products that enhanced fiber-matrix interfacial bonding and fiber dispersion. The enhanced fiber bridging capacity due to increased bond strength contributed to superior strain hardening and crack control. The material achieved a balance between strength and sustainability, showing high toughness and controlled micro-cracking with crack widths less than 100 µm. This advancement enables the use of construction waste and alternative binders to reduce carbon footprint while enhancing material ductility, supporting a green and durable alternative for future infrastructure applications.

Overall, research in sustainable construction has focused on improving structural performance while reducing environmental impact through innovative materials and joint designs. Advances in timber connections have shown that modifying bore holes with adhesive bulbs in glued-in steel rod joints enhances load capacity, ductility, and residual strength without altering the rods themselves. Similarly, optimized dowel detailing in precast concrete beam-to-column connections improves seismic performance, energy dissipation, and post-peak behavior, demonstrating that careful reinforcement design can make precast systems comparable to monolithic joints. On the materials side, several studies have explored the partial replacement of natural aggregates with industrial or post-consumer waste. Waste foundry sand, recycled crushed glass, recycled ceramic aggregates, foam glass, and recycled fine concrete aggregates can replace conventional sand or coarse aggregates in concrete and mortar, generally achieving optimal performance at moderate substitution levels (around 20–40%). These substitutions enhance compressive, tensile, and flexural strengths, improve durability, reduce permeability, and promote sustainability by diverting waste from landfills. Innovative cementitious systems incorporating limestone calcined clay and recycled aggregates further improve ductility, crack control, and fiber-matrix bonding, achieving a balance between mechanical performance and environmental responsibility. In a nutshell, the integration of optimized joint design, alternative aggregates, and engineered binders supports a circular, low-carbon approach to construction while maintaining structural integrity, durability, and seismic resilience.

Green manufacturing

Mani et al. (2014) have examined sustainability characterization for additive manufacturing (AM) with the purpose of identifying environmental impacts, performance gaps, and standardization requirements to benchmark AM processes against traditional manufacturing. The study focused on materials such as metals, polymers, ceramics, and composites in the form of powders, wires, and liquids, employed across the seven ASTM F42-categorized AM technologies, with equipment including polymer laser sintering platforms like Sinterstation HiQ + HS and EOSINT P 390, selective laser sintering (SLS) machines, and direct metal deposition (DMD) systems. Process parameters assessed involved energy consumption (time-dependent, geometry-dependent, and Z-height-dependent), material efficiency, embodied energy, carbon footprint, and life-cycle resource utilization. Test results demonstrated that AM significantly reduces raw material usage and waste compared to machining and casting, enables lightweight part design with high strength-to-weight ratios, and minimizes supply chain and transportation needs. Energy assessments revealed that although AM can have higher machine energy consumption in some processes, overall environmental impact is reduced when accounting for material savings, repair/remanufacturing potential, and design optimization capabilities such as topology optimization. The major technical finding emphasized the importance of developing standardized sustainability indicators and measurement frameworks for AM, highlighting that

sustainable design practices, material selection, and process optimization are essential to maximize AM's environmental and economic benefits. By enabling reduced waste, resource efficiency, and on-demand production, AM demonstrates significant promise as a sustainable manufacturing paradigm when supported by standardization and performance benchmarking.

Thiede et al. (2016) have explored sustainable additive manufacturing with a focus on energy demand along the process chain. The purpose was to identify energy consumption patterns of AM compared to conventional processes and propose improvements for energy efficiency. The study examined polymer-based selective laser sintering (SLS) and fused deposition modeling (FDM) systems, considering materials such as polyamide powders and polymer filaments, while equipment used included commercial laser sintering platforms and extrusion-based 3D printers. Process parameters analyzed covered build chamber temperature (ranging up to 170 °C for SLS), laser power (tens of watts), build time (several hours per job), and part geometry affecting Z-height energy demand. Results indicated that SLS exhibits significantly higher specific energy consumption (up to 50–60 MJ/kg of part) compared to conventional polymer processing such as injection molding, while FDM showed lower energy demand but limited part quality and productivity. The research highlighted that preheating and idle phases contribute substantially to total energy use, with more than 30% of the energy consumed not directly contributing to material consolidation. A major technical insight was that system design improvements, heat recovery mechanisms, and optimized process scheduling could substantially reduce energy intensity. By mapping the energy footprint of AM technologies, the study demonstrated that although AM currently lags behind traditional methods in energy efficiency, targeted process and equipment optimization can improve sustainability and align AM with broader environmental goals.

Kurdari et al. (2021) have investigated metal additive manufacturing (AM) recycling and reuse strategies with the purpose of evaluating how metal powders and parts can be efficiently recycled to enhance sustainability and reduce production costs. The study focused on commonly used AM materials such as titanium, aluminum, nickel-based alloys, and stainless steel, processed primarily through selective laser melting (SLM), electron beam melting (EBM), and direct energy deposition (DED) systems. Equipment highlighted included high-power laser systems, electron beam platforms, and powder handling and sieving machines used for material recovery. Process parameters assessed encompassed powder recycling cycles (up to 14 reuse cycles), oxygen content thresholds (typically < 0.2 wt%), particle size distribution changes (15–45 µm range for SLM), and mechanical property retention across multiple builds. Test results revealed that while powders can generally be reused multiple times with minimal property degradation, accumulation of contaminants such as oxygen and nitrogen, as well as morphological changes like increased satellite particles, can negatively affect flowability and part density. Key technical findings showed that controlled sieving, blending with virgin powders, and monitoring of chemical composition are essential for ensuring consistent quality in recycled materials. The research demonstrated that recycling not only reduces raw material demand and cost by up to 30% but also significantly cuts environmental impacts by lowering energy and resource use associated with virgin powder production. By advancing powder recycling protocols and establishing standards for reuse, metal AM can achieve greater economic viability and environmental sustainability without compromising mechanical performance.

Korkmaz et al. (2023) have examined the implementation of green cooling and lubrication strategies in metal cutting industries, with the purpose of identifying sustainable alternatives to conventional practices that reduce environmental harm, occupational hazards, and excessive resource consumption. The study focuses on a review-based methodology without using experimental materials or lab equipment, instead leveraging literature sources from databases like Elsevier, Springer, and Scopus. It explores techniques such as dry machining, Minimum Quantity Lubrication (MQL), Nano-MQL, and cryogenic cooling, assessing their effectiveness in terms of energy consumption, tool life, carbon footprint, and health safety. Process parameters include cutting fluid type, cooling conditions, and lubrication technique, though ranges are qualitatively rather than quantitatively discussed. The results show that advanced green machining strategies not only enhance process efficiency and reduce waste but also significantly lower energy usage and emissions, especially when compared to conventional flood cooling systems. A major technical takeaway is that these green techniques can transform machining processes into more eco-friendly, health-conscious, and cost-effective systems, contributing toward sustainable industrial development by minimizing toxic discharge and optimizing resource use.

Siahsarani et al. (2025) have investigated sustainable and efficient cooling methods in titanium milling for dental applications using supercritical CO_2 combined with minimum quantity lubrication ($scCO_2 + MQL$). The purpose of the study was to address the machining challenges associated with Grade 4 titanium, commonly used in dental implants, due to its low thermal conductivity and tendency to cause rapid tool wear and surface defects. Materials used included Grade 4 titanium discs and solid carbide ball nose end mills, with surface lubrication via Survos Standard MQL oil and emulsion-based coolant AquaTec 7655. Equipment included a five-axis CNC milling machine (GF Mikron MILL S 400 U) equipped with Fusion Coolant Systems Pure-Cut + ® for

applying $scCO_2$ at 105 bar (10 kg/h flow rate) with MQL oil at 2 ml/min. The process parameters covered varying cutting speeds and feeds, comparing the $scCO_2$ + MQL system against traditional flood cooling. Test results demonstrated that $scCO_2$ + MQL reduced tool flank wear by over 190% and increased material removal rate by approximately 68%, with significant improvement in surface integrity and dimensional precision in complex dental geometries. The major technical advancement lies in the Joule–Thomson effect-induced CO_2 snow formation, which rapidly dissipates heat and enhances tool life, making $scCO_2$ + MQL a superior, eco-friendly alternative for precision machining in biomedical sectors, particularly for complex dental implant manufacturing.

Tembo et al. (2025) have experimented with propionic acid as a green lixiviant for the recovery of cathode metals from lithium-ion battery (LIB) materials, aiming to provide a sustainable alternative to conventional leaching agents in battery recycling. The study examined both virgin cathode powder (VCP) and spent cathode powder (SCP) derived from $LiNiCoMnO_2$ (NCM 523) materials. Materials used included commercially sourced cathode powders, propionic acid (55–58%), hydrogen peroxide (H_2O_2), and deionized water, with leaching solutions freshly prepared. Equipment employed comprised inductively coupled plasma mass spectrometry (ICP-MS) for metal analysis and scanning electron microscopy (SEM) for material characterization. Process parameters evaluated included temperature (ambient), acid concentration, leaching time, and metal recovery percentages. Results showed high metal recoveries from SCP—92.9% Co, 87.4% Li, 92.7% Mn, and 94.0% Ni—with only minor differences compared to VCP. Kinetic analysis identified the leaching mechanism as chemically controlled, supported by activation energy values ranging from 53.38 to 63.98 kJ/mol across the metals. The major technical advancement is the confirmation of propionic acid's effectiveness as a monoprotic, low-toxicity, and recyclable organic acid for hydrometallurgical recycling, offering comparable recovery performance to stronger inorganic acids but with significantly lower environmental impact. This approach supports closed-loop recycling for LIBs and reduces the ecological footprint of battery waste management.

Trinca et al. (2025) have investigated a hybrid hydro-pyrometallurgical process for sustainable steel production from low-grade iron ores. The study aims to develop a novel industrial-scale system with net-negative CO_2 emissions by integrating oxalic acid-based hydrometallurgy and hydrogen-assisted pyro-reduction. Materials used include low-grade iron ores, oxalic acid (as the leaching agent), carbon dioxide (for oxalic acid regeneration), and hydrogen (for final reduction), while equipment involved includes electrolyzers, photoreactors, and electric arc furnaces. Key process parameters explored include oxalic acid concentration for dissolution, light intensity for photoreduction, and hydrogen/carbon monoxide ratios for pyro-reduction, with optimization modeled in Aspen Plus software. Experimental and simulation results showed that iron production is feasible at a levelized cost of 1093.32 $/tSTEEL using renewable energy at 30 $/MWh. The photoreduction yielded high selectivity in converting Fe^{3+} to Fe^{2+} and the final step efficiently produced metallic iron. A notable finding is the effective regeneration of oxalic acid via electrochemical CO_2 reduction, enhancing the process's circularity. Additionally, comparison with conventional hydrogen-DRI technologies indicated improved environmental performance, though energy demand remains a constraint pending future advancements in electrolyzer efficiency. This integrated process enables cost-competitive and environmentally superior steel production from previously underutilized low-grade ores.

Wei et al. (2025) have developed a novel metal/biopolymer composite filament for extrusion-based additive manufacturing using CuSn10 as an example, targeting the production of metal components through material extrusion (MEX) with a sustainable binder system. The study aimed to enhance metal powder content while ensuring printability and effective post-processing. CuSn10 metal powder was used at 65 vol% concentration, mixed with two different biopolymer binder systems—first, polylactic acid (PLA) and acetyl tributyl citrate (ATBC); second, PLA, butenediol vinyl alcohol copolymer (BVOH), and ATBC. Feedstock characterization and fabrication included extrusion into filaments, printing green parts using MEX, and subsequent thermal debinding and sintering in an open-air muffle furnace. Process parameters included binder ratios (optimal at PLA with 10 vol% ATBC and PLA/BVOH with 5 vol% ATBC), extrusion temperatures (~ 190–210 °C), and sintering profiles suitable for copper alloys. Results revealed that the plasticizer enhanced filament flowability but slightly reduced surface quality, while the addition of BVOH improved both viscosity and smoothness. Successful sintering produced dense CuSn10 parts with improved mechanical strength and low porosity, demonstrating the viability of bio-based binders for environmentally friendly metal additive manufacturing. The research illustrates a scalable and eco-conscious pathway to fabricate complex metal parts for applications in industries like aerospace and healthcare.

Overall, research highlights the growing role of additive manufacturing (AM) and green technologies in sustainable production. AM significantly reduces material waste, enables lightweight and optimized part designs, and allows on-demand, localized production, cutting supply chain emissions. While energy consumption can be higher in some AM processes, efficiency gains are achieved through design optimization, topology optimization, and repair/

remanufacturing potential. Material recycling, particularly for metals, enhances sustainability by reducing raw material demand and maintaining part quality through controlled sieving, blending, and chemical monitoring. Energy efficiency remains a key challenge, especially in polymer-based AM, but strategies such as improved system design, heat recovery, and optimized process scheduling can mitigate excess energy use. Green cooling and lubrication techniques, including minimum quantity lubrication, cryogenic cooling, and supercritical CO_2 methods, improve machining efficiency, reduce tool wear, lower emissions, and enhance workplace safety. Emerging innovations in metal AM integrate circular economy principles, AI-driven process control, digital twins, and advanced powder recyclability, enabling resource conservation and closed-loop manufacturing. Sustainable materials, including biopolymer-metal composites for extrusion-based AM, further reduce environmental impact while maintaining mechanical performance. Parallel developments in eco-friendly metal recovery, battery recycling using organic acids, and hybrid hydrometallurgical-pyrometallurgical steel production demonstrate broader industrial potential for low-carbon, resource-efficient, and circular manufacturing systems. Nutshell, these advances illustrate that combining AM, green machining, sustainable material selection, and process optimization can drive a more environmentally conscious and economically efficient manufacturing paradigm.

Manufacturing innovation

Shahzad et al. (2022) have investigated the adoption of green innovation technology to accelerate sustainable development among the manufacturing industry. The study aimed to understand the behavioral and organizational factors influencing green innovation adoption (GIA) in Pakistani manufacturing firms, applying an integrated model based on the Unified Theory of Acceptance and Use of Technology (UTAUT). A total of 516 survey responses were collected and analyzed using structural equation modelling (SEM) and artificial neural network (ANN) techniques. No physical materials or lab equipment were used; however, statistical tools like Smart PLS and ANN modelling platforms served as methodological instruments. Process parameters involved examining constructs such as performance expectancy, effort expectancy, hedonic motivation, facilitating conditions, innovation cost, and social influence, with their predictive power on green behavioral intention (GBI). The results revealed that all constructs significantly influenced GBI, which in turn positively mediated green innovation adoption. Among the constructs, green facilitating conditions showed the highest relative importance in the ANN analysis. Additionally, organizational size moderated the relationship between behavioral intention and adoption, showing differences across SMEs and large enterprises. The research demonstrated that strategic behavioral and environmental conditions significantly dictate the success of green innovation in achieving sustainable development goals within manufacturing, suggesting that fostering these internal and external conditions can robustly support the green transformation of industrial operations.

Sohns et al. (2023) have developed a framework for green business process management (Green BPM) for business sustainability in manufacturing SMEs, aiming to assess and improve how small and medium-sized enterprises manage their processes with environmental sustainability in mind. The study integrates the Green BPM readiness model with the Business Sustainability Typology (BST) to create a maturity model applied through multiple case studies across ten German SMEs. No physical materials or equipment were used, as the research was conceptual and case-based. The model evaluates six key Green BPM factors: green attitude, strategy, governance, modeling, monitoring, and optimization. Process parameters include organizational behavior toward sustainability, managerial structures, and process monitoring practices. Test results revealed variability in maturity levels, with most SMEs showing strong green attitude but struggling with strategic execution and governance alignment. A significant technical aspect is the demonstration of how process-centric maturity assessment can guide targeted sustainability improvements. The study concludes that while SMEs are aware and supportive of green practices, they often lack systematic implementation pathways, which can be addressed through structured Green BPM frameworks that help translate intent into measurable process improvements.

Bendig et al. (2023) have performed a systematic review on green manufacturing frameworks, aiming to construct a structured and comprehensive model that categorizes the influences, elements, and results of green manufacturing research across disciplines. The study examined 6000 publications from multiple databases, narrowed down to 189 through strict inclusion criteria, without using experimental materials or equipment but relying heavily on bibliometric and content analysis tools such as MAXQDA. The process involved classifying 290 codes into thematic segments and categories—spanning performance, people, and interdisciplinary dimensions. The analysis used descriptive statistics, co-occurrence mapping, and concept-centric synthesis to identify major trends, with process parameters defined by frequency and co-linkages of keywords in academic literature. Results highlighted an increasing scholarly interest since 2013, with prominent themes including sustainability, lean integration, implementation challenges, and efficiency strategies. The major technical contribution is the development of a detailed green manufacturing framework that visualizes the evolution of the field, identifies research gaps,

and proposes 20 targeted future research questions, offering strategic direction to scholars and practitioners.

Haleem et al. (2023) have reviewed green manufacturing strategies for achieving sustainability, aiming to redefine industrial practices using the 5R (repair, reuse, refurbish, rebuild, recycle) philosophy for environmental and economic gain. The study is qualitative, drawing from literature across platforms like Scopus, ScienceDirect, and ResearchGate. Materials and equipment are discussed conceptually including energy-efficient motors, green computing devices, and cleantech textiles. Process parameters encompass energy efficiency, emission reduction, waste recycling, water conservation, and lifecycle management. No specific numerical tests are conducted; however, the technical focus is on tools such as green dashboards, AHP, WLC models, circular economy practices, and green software systems. The review concludes that integrating smart technologies like IoT and big data analytics with green initiatives enables firms to minimize ecological footprints, boost resource efficiency, and drive sustainable innovation, ultimately fostering a green corporate culture and achieving long-term environmental compliance.

Li et al. (2023) have analyzed the role of green production and green technology in promoting sustainability through the mediating effects of waste reduction and energy use, aiming to quantify their interrelations within the textile manufacturing industry. The study utilized data collected via surveys and applied partial least squares structural equation modeling (PLS-SEM) for evaluation, without the use of traditional experimental equipment but leveraging statistical modeling tools. The materials of focus include green technologies and production practices implemented in textile firms, with process parameters encompassing waste generation, energy consumption, and adoption rates of eco-friendly methods. The findings reveal that green production and green technology significantly enhance sustainability by reducing both waste and energy usage, which in turn serve as key mediators in improving overall environmental performance. A critical technical insight is that companies adopting such practices benefit not only from lower ecological footprints but also gain competitive advantage and cost savings. The study offers a structured theoretical framework based on Diffusion of Innovation Theory, Resource-Based View, and Circular Economy Theory, demonstrating that integrating green technologies in manufacturing directly contributes to sustainable industrial transformation.

Bilancia et al. (2025) have reviewed the role of lean practices in achieving sustainability and circular economy within the luxury fashion industry, with the objective of identifying how lean methodologies can reduce waste, improve environmental outcomes, and promote circular production systems. The study performed a systematic literature review analyzing academic and industrial literature on lean applications such as Lean Six Sigma, Value Stream Mapping (VSM), 5S, Kanban, Kaizen, Single Minute Exchange of Die (SMED), and visual management across luxury clothing production. No physical materials or equipment were used as the research was conceptual and data-driven. Process parameters focused on production efficiency, waste generation, and cycle time improvements within low-volume high-variety (LVHV) production models. Results highlighted that lean practices effectively reduce energy use, solid waste, and emissions while enhancing product quality and operational efficiency. Lean integration with Industry 4.0 technologies such as barcode tracking and real-time data analytics further supports traceability, eco-efficiency, and process transparency. The major technical takeaway is the positive environmental performance of lean when aligned with sustainable development goals and circular economy principles, despite limited implementation in luxury segments. This review revealed significant opportunities to scale lean practices in high-quality and customizable fashion sectors, offering dual gains in environmental sustainability and competitive advantage.

Costa et al. (2024) have developed a comprehensive model for integrating Industry 4.0 and Lean Manufacturing to support a sustainable green transition. The objective of the study was to bridge the gap in existing literature by proposing a unified framework that synthesizes Lean Manufacturing (LM), Industry 4.0 (I4.0), and Green Paradigm (GP) principles to enhance environmental sustainability in industrial settings. The methodology included a systematic literature review and the ISM-MICMAC analytical method, with model validation performed through industry interviews rather than experimental equipment or materials. Process variables identified and analyzed included 27 critical elements spanning digital technologies (e.g., IoT, AI, big data), lean tools (e.g., Just-in-Time, Total Quality Management), and green practices (e.g., life-cycle assessments, eco-design, and reverse logistics). The model illustrated transitive relationships between these variables, identifying core enablers and dependencies. Results showed that the integration of I4.0 technologies and LM practices not only improves operational efficiency and resource utilization but also significantly supports sustainable goals by reducing waste, emissions, and energy consumption. The model provides companies with a strategic roadmap for green transition, enabling practical implementation of sustainable innovations that align with both environmental objectives and competitive industrial performance.

Rame et al. (2024) have examined the emerging trends and challenges in Industry 5.0 and its integration with sustainability for a green future. This study aimed to understand how Industry 5.0 technologies—characterized by the synergy between human intelligence and advanced digital systems—can be harnessed to support sustainable development across sectors such as manufacturing, energy, and urban

planning. Conducted through a systematic literature review, the study relied on academic databases like ScienceDirect and Scopus without using physical materials or lab-based equipment, focusing instead on thematic data synthesis. The analysis was structured around process variables including technological integration, human–machine collaboration, circular economy practices, and the environmental impacts of advanced automation. The paper discussed how technologies like artificial intelligence and the Internet of Things contribute to resource efficiency, waste reduction, and circular production systems. It also explored barriers such as the economic-environmental trade-offs and implementation constraints of these technologies in developing economies. By mapping innovation trends and aligning them with sustainability frameworks, the study concluded that Industry 5.0 holds transformative potential to bridge environmental imperatives with technological advancements, creating resilient, human-centered industrial systems that drive both ecological balance and economic growth.

Sarker and Bartok (2024) have conducted a bibliometric review on global trends of green manufacturing research in the textile industry, aiming to map the evolution, thematic clusters, and collaboration networks within this domain. The study analyzed 523 Scopus-indexed publications using PRISMA 2020 guidelines and VOS viewer software, without employing experimental materials or lab equipment. It systematically visualized publication trends, co-authorship networks, and citation patterns, focusing on process parameters such as research frequency, keyword co-occurrence, and topical clustering. Key findings identified five major thematic clusters: life cycle assessment, lean approaches, sustainability, cleaner production, and waste management, with China emerging as the leading contributor to collaborative research. A significant technical insight is that despite a sharp rise in interest since 2013, the field still requires more integration of circular and digital technologies and broader data sources to enhance sector-wide transformation. The study emphasizes the urgent need for actionable green practices in the textile sector, driven by the industry's heavy environmental burden, and provides a roadmap for future scholarly work that encourages sustainable innovations and policy intervention.

Boumsisse et al. (2025) have investigated the integration of Industry 5.0 technologies into Green Lean Six Sigma (GLSS) to enhance sustainable and efficient manufacturing processes. The purpose of this study was to create a structured framework that maps Industry 5.0 technologies onto the DMAIC (Define, Measure, Analyse, Improve, Control) cycle of GLSS, aiming to bridge the gap between advanced technology and operational sustainability. Through a documentary research approach, the study relied on an extensive literature review, using databases like Scopus, IEEE Xplore, and Web of Science, with keywords such as "Industry 5.0," "DMAIC", and "sustainability." Although no physical experiments were performed, the analysis emphasized integrating technologies like artificial intelligence (AI), digital twins, the Internet of Everything (IoE), and advanced data analytics at each DMAIC stage. AI was highlighted for improving decision-making and reducing operational errors; IoE for enhancing real-time monitoring and minimizing communication bottlenecks; and digital twins for simulating and optimizing product lifecycle impacts. These technologies were shown to support sustainable metrics such as reduced energy consumption, waste minimization, and improved process efficiency. The research proposed a novel theoretical framework that guides manufacturers in deploying these technologies across each phase of DMAIC, offering detailed recommendations for practical implementation. The findings underline the synergistic potential of combining GLSS and Industry 5.0 for achieving ecological and operational excellence, laying the groundwork for future empirical validations in industrial environments.

Goh et al. (2025) have developed a multi-objective optimisation framework for designing biowaste-based green hydrogen supply chains (GHSC) with integrated economic, environmental, and safety considerations. The study addresses the growing need for sustainable hydrogen production by proposing a mixed-integer programming (MIP) model applied to palm oil industrial wastes in Malaysia. Materials considered include palm oil mill residues such as empty fruit bunches and palm oil mill effluent (POME), which serve as biomass feedstock, while technological components involve biohydrogen production systems and solar-powered electrolysis units. Process parameters evaluated span feedstock allocation, facility location, hydrogen storage technologies, and energy sources, including solar integration scenarios. The model evaluates scenarios based on profitability, carbon emissions, and safety risks, applying tools like GAMS with CPLEX for computation. Results demonstrate that multi-centralized biowaste hubs and hybrid technology pathways significantly improve supply chain efficiency and emissions reduction. Sensitivity analysis reveals that solar-powered electrolysis becomes economically viable when the hydrogen price exceeds USD 5.36/kg. Notably, safety risks—often overlooked—are embedded into the decision-making structure, enabling a comprehensive and resilient design strategy. The study presents a scalable decision-support system that enhances local resource valorisation, reduces dependence on fossil fuels, and ensures the feasibility of green hydrogen deployment across diverse geographic and industrial contexts.

Liu et al. (2025) have explored the green transformation of China's traditional manufacturing industry driven by the digital economy using a system dynamics approach. The study aimed to understand how digital technologies, policy frameworks, and organizational readiness collectively

influence green development in traditional manufacturing sectors under China's dual-carbon goals. Utilizing the Technology-Organization-Environment (TOE) framework and system dynamics modeling via Vensim software, the authors conducted simulations based on empirical data from 2015 to 2020. Although no direct materials or physical equipment were used, the simulation process incorporated policy input variables, innovation capacities, and emission control metrics. Key process parameters included digital innovation, environmental regulation, and technological standardization, which were simulated across various scenarios. Results demonstrated that environmental policies, digital technology application, and organizational capabilities can collectively boost green total factor productivity while reducing CO_2 emissions. The study highlighted that digital transformation not only enhances operational efficiency but also catalyzes green initiatives by enabling real-time data integration, policy responsiveness, and strategic alignment with sustainability goals. These insights contribute valuable guidance for policymakers and industrial leaders seeking to accelerate the ecological modernization of traditional manufacturing.

Overall, research on manufacturing innovation emphasizes the integration of green practices, digital technologies, and lean methodologies to achieve sustainable industrial development. Studies highlight that behavioral, organizational, and environmental factors strongly influence the adoption of green innovations, with facilitating conditions and managerial support being key drivers. Green business process management and maturity models show that while SMEs often exhibit positive environmental attitudes, they struggle with systematic implementation, underscoring the need for structured frameworks to translate intent into measurable outcomes. Frameworks combining Industry 4.0, Industry 5.0, and lean principles demonstrate that digital tools, AI, IoT, and advanced analytics enhance operational efficiency, resource utilization, and environmental performance. Similarly, green technologies in production, waste reduction, energy efficiency, and circular economy practices contribute to reduced emissions, improved resource use, and competitive advantage. Bibliometric and systematic reviews reveal growing global interest in green manufacturing, lean integration, and sustainability practices, while also identifying research gaps in digital adoption and circular economy implementation. Sector-specific applications, such as textile manufacturing, luxury fashion, and biohydrogen supply chains, show that tailored approaches combining technological, managerial, and process innovations can simultaneously drive economic, environmental, and safety outcomes. Overall, the convergence of behavioral strategies, advanced technologies, and structured process management forms a comprehensive pathway for sustainable industrial transformation, fostering green competitiveness, efficiency, and long-term ecological benefits.

Water treatment

Shahid et al. (2021) have reviewed the application of electrocoagulation in wastewater treatment, aiming to consolidate findings on operational mechanisms, influencing parameters, and performance efficiency of this electrochemical technique. The study examines sacrificial anodes of aluminium and iron as primary materials, with wastewater matrices including textile effluents, tannery wastewater, dye-laden streams, and municipal sewage. Equipment used includes batch and continuous electrocoagulation cells fitted with DC power supplies, electrodes, and settling tanks. Process parameters vary with current density (5–50 mA/cm^2), electrolysis times from 5 to 60 min, electrode spacing between 1 and 5 cm, and initial pH levels ranging from acidic to alkaline (3–10). Results consistently show pollutant removal efficiencies above 90% for turbidity, suspended solids, COD, BOD, and heavy metals, with energy consumption rates of 0.5–5 kWh/m^3 depending on conditions. The major technical insight is that electrode material, current density, and water chemistry strongly influence floc formation, pollutant aggregation, and energy demand, with hybrid systems (EC combined with oxidation or membranes) offering enhanced treatment efficiency. The study emphasizes electrocoagulation as a robust, versatile, and scalable solution for treating complex wastewaters, with optimization of operational parameters being key to cost-effective implementation.

Zheng et al. (2023) have examined the integration of microalgae cultivation and wastewater treatment to address sustainable resource recovery and environmental remediation, focusing on coupling pollutant removal with biomass generation. The study highlights the use of municipal and industrial wastewater as the primary medium, where nutrients such as nitrogen and phosphorus serve as feed for algal growth. Cultivation equipment includes open raceway ponds and photobioreactors, while operational parameters span light intensities between 50 and 200 μmol/m^2·s, hydraulic retention times from 2 to 10 days, and temperature ranges of 20–35 °C. Test results demonstrate that microalgae-based systems achieve nutrient removal efficiencies of over 90% for nitrogen and phosphorus, alongside reductions in chemical oxygen demand by 60–80%. Biomass yields are reported between 0.3 and 0.8 g/L, suitable for downstream applications such as biofuels, animal feed, and biofertilizers. A major technical advancement is the potential of integrating anaerobic digestion and bioelectrochemical systems to enhance energy recovery and achieve a circular wastewater–bioenergy nexus. The findings emphasize that microalgae–wastewater integration provides a dual benefit of water quality improvement and bioresource recovery, making it a promising pathway for sustainable wastewater management and renewable energy production.

Awokoya et al. (2024) have synthesized a trimethoprim vanillin anchored conjugate imprinted polymer (CIA) for the selective removal of bromocresol green (BCG) and malachite green (MG) from aqueous media. The study was conducted to address the environmental and health hazards posed by these toxic dyes commonly found in textile wastewater. The material developed was based on molecular imprinting technology using bulk polymerization, incorporating trimethoprim-vanillin (TMP-V) as the ligand and styrene as the functional monomer. Characterization of the synthesized polymer was performed using X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), and scanning electron microscopy (SEM). Batch adsorption tests evaluated performance under varying parameters including contact time (up to 90 min), dye concentration (10–60 mg/L), pH (3–11), and temperature (25–65 °C). The imprinted polymer achieved maximum removal efficiencies of 99.27% for BCG and 98.99% for MG, significantly outperforming the non-imprinted polymer which achieved only 51.52% and 62.90%, respectively. Kinetic modeling confirmed pseudo-second-order behavior, while isotherm data aligned well with Langmuir models, indicating monolayer adsorption. Thermodynamic analysis revealed BCG adsorption was non-spontaneous, whereas MG adsorption was spontaneous, with both processes being endothermic. The polymer also showed robust reusability, retaining high efficiency over five cycles and demonstrating high performance in real textile effluent samples with 85.5–87.5% dye removal. The combination of ligand-based imprinting with enhanced cavity formation due to π–π interactions led to high affinity and selectivity, confirming the synthesized CIA as a promising and sustainable adsorbent for wastewater treatment applications.

Shamshad and Rehman (2024) have investigated the emerging role of nanotechnology in wastewater treatment, aiming to evaluate how nanomaterials can enhance contaminant removal efficiency, reduce energy consumption, and support sustainable water management. The study reviews nanomaterials including carbon nanotubes, metal oxide nanoparticles (TiO_2, ZnO, Fe_3O_4), and nanocomposites, highlighting their adsorption, photocatalysis, and antimicrobial properties. Equipment discussed in the reviewed works includes membrane bioreactors, photocatalytic reactors, and adsorption columns integrated with nanomaterials. Process parameters reported span pH ranges of 3–11, contact times from minutes to several hours, nanoparticle dosages typically between 0.1 and 1.0 g/L, and light wavelengths for photocatalysis in the UV–visible spectrum. Results indicate that nanomaterials achieve removal efficiencies above 90% for heavy metals (Pb^{2+}, Cr^{6+}), dyes, pharmaceuticals, and microbial pathogens, outperforming conventional adsorbents and catalysts. A key technical aspect lies in the regeneration and reusability of nanomaterials, which can retain high efficiency after multiple cycles, though challenges remain regarding cost, large-scale application, and environmental safety of nano waste. The synthesis highlights that nanotechnology provides transformative potential for wastewater treatment by offering multifunctional, high-performance materials that can be tailored for specific contaminants, paving the way for greener and more resilient water purification systems.

Kandasamy et al. (2025) have synthesized a green ZnO/NiO heterojunction nanocomposite for solar energy conversion, photocatalysis, and energy storage applications, aiming to provide an eco-friendly multifunctional nanomaterial with enhanced optoelectronic and electrochemical performance. The study utilized a *Syzygium cumini* leaf extract in a green synthesis approach to fabricate the ZnO/NiO nanocomposite (NC), targeting reduced environmental impact and cost. Materials involved were zinc and nickel precursors, while the *Syzygium cumini* leaf extract served as a bio-reducing and stabilizing agent. Equipment employed includes UV–vis spectroscopy, XRD, FTIR, FESEM, and EDX for characterization, with photovoltaic testing for DSSC evaluation and electrochemical measurements for supercapacitor testing. Process parameters included the use of ZnO and NiO precursors in optimal ratios, pH adjustment during synthesis, and annealing under controlled conditions; specific ranges for temperature and pH were not detailed but involved typical mild conditions consistent with green chemistry. The test results showed that the DSSC photoanode using the ZnO/NiO NC achieved a power conversion efficiency (PCE) of 2.09%, outperforming pristine ZnO (1.63%). The photocatalytic degradation of methylene blue dye reached 99% with the NC compared to 88% with pure ZnO, attributed to enhanced charge separation and suppressed recombination. In supercapacitor applications, the NC exhibited a specific capacitance of 262.50 F/g and cyclic stability of 77.84% over 1000 cycles. The composite's performance across these domains indicates its strong potential for integrated applications in renewable energy and environmental remediation technologies.

Overall, research in wastewater treatment and environmental remediation reveals a convergence of electrochemical, biological, adsorption-based, and nanomaterial-driven strategies aimed at enhancing removal efficiency, energy recovery, and sustainability. Electrocoagulation using aluminium and iron electrodes demonstrates pollutant removal above 90% for turbidity, COD, BOD, and heavy metals, with process performance strongly governed by current density between 5 and 50 mA/cm^2, electrode spacing of 1–5 cm, and water chemistry, while hybrid integration with oxidation or membranes further improves efficiency and scalability. Microalgae wastewater systems achieve dual benefits of nutrient removal exceeding 90% for nitrogen and phosphorus and COD reduction of 60–80%, while simultaneously

generating biomass yields up to 0.8 g/L for conversion into biofuels, fertilizers, or feed, with potential gains when coupled with anaerobic digestion and bio-electrochemical energy recovery. Molecularly imprinted polymers tailored through ligand-specific cavities deliver selective dye removal efficiencies above 98%, with pseudo-second-order adsorption kinetics and Langmuir monolayer behavior confirming optimized surface interactions and robust recyclability in real textile effluents. Nanotechnology-based interventions using carbon nanotubes, TiO_2, ZnO, Fe_3O_4, and nanocomposites consistently achieve > 90% removal of dyes, heavy metals, pharmaceuticals, and microbes, with performance influenced by nanoparticle dose (0.1–1.0 g/L), pH (3–11), and UV–visible activation, while demonstrating regeneration potential, albeit with scale-up and nano-waste challenges. Green-synthesized ZnO/NiO heterojunction nanocomposites provide multifunctionality, achieving 99% photocatalytic dye degradation, 2.09% DSSC efficiency, and supercapacitor capacitance of 262.5 F/g with strong cycle stability, highlighting the integration of photocatalysis, energy conversion, and storage into a single eco-friendly platform. Taken together, these approaches underscore that future wastewater management and environmental remediation will rely on optimizing electrochemical and biological processes, tailoring adsorbent selectivity, advancing nanomaterial multifunctionality, and adopting green synthesis pathways to deliver scalable, efficient, and sustainable solutions.

Circular resource use

Alam et al. (2023) have analyzed drivers of Agriculture 4.0 implementation in emerging economies focusing on sustainability and food security, particularly in the context of Bangladesh. The purpose was to prioritize and model the interrelationships among implementation drivers using Pareto analysis, fuzzy Total Interpretive Structural Modelling (TISM), and fuzzy MICMAC. Materials include automation tools like IoT, robotics, AR for training, cloud computing, AI, and BDAs; no experimental equipment was employed. The process parameters encompassed policy, financial support, training, and digital infrastructure, ranked across 18 initial drivers, later reduced to 13 for modelling. Findings highlight that automation, AR-based farmer training, and financial incentives for storage infrastructure are the most critical drivers. Fuzzy MICMAC analysis positioned these as highly influential variables. The study enables strategic policymaking for integrating digital agriculture technologies, improving food system resilience, and achieving the SDGs in resource-constrained regions.

Alqadi et al. (2023) have conducted a comparative analysis on the life cycle energy and greenhouse gas emissions of natural and artificial stone during the manufacturing phase, aiming to assess environmental impacts in the context of Palestine's stone-cutting industry. The study employed natural limestone and artificial stone composed of cement, sand, and aggregates, with data gathered from quarry site visits, workshops, and interviews. Equipment and techniques used include life cycle assessment (LCA) tools to analyze energy inputs and emissions, though specific LCA software is not disclosed. The primary process parameters compared were energy consumption per square meter and associated GHG emissions. Natural stone production consumed approximately 120 MJ/m^2 and emitted more GHGs, whereas artificial stone manufacturing used about 70.71 MJ/m^2, indicating a lower environmental impact. Key findings highlight that the artificial alternative is more sustainable under the studied socio-political and logistical constraints of Hebron, where transportation inefficiencies and energy import dependencies exacerbate natural stone's carbon footprint. The study suggests environmental performance can be improved by relocating workshops closer to raw material sources, adopting green energy, and using electric vehicles, showcasing how spatial planning and energy strategy can mitigate ecological damage in regional construction material supply chains.

Debnath et al. (2023) have analyzed the critical success factors (CSFs) for implementing green supply chain management (GSCM) in the apparel manufacturing industry of emerging economies. The study aimed to identify and evaluate interrelationships among CSFs that influence the adoption of GSCM, supporting sustainable development goals (SDGs) in Bangladesh. A hybrid multi-criteria decision-making (MCDM) approach combining grey theory and the Decision-Making Trial and Evaluation Laboratory (DEMATEL) method was employed. Sixteen CSFs were identified through literature review and expert validation, with no physical equipment used but a detailed analytical framework involving expert responses and matrix modeling. Key process parameters were the cause-effect influences among the CSFs, with "economic and tax benefits," "demand from buyers," and "government rules and regulations" emerging as the most influential drivers. Among these, "supplier training and cooperation" was found to be most influenced. The analysis revealed that policy incentives, buyer expectations, and regulatory frameworks are foundational to fostering GSCM in the apparel sector. By establishing causal relationships among factors, the study provides actionable insights for managers and policymakers to develop effective strategies for GSCM adoption that align with both sustainability objectives and competitive manufacturing.

Fiorello et al. (2023) have proposed a framework for a smart lean green production paradigm to improve operational performance, focusing on integrating lean management, sustainability, and Industry 4.0 technologies. The purpose is to guide companies, particularly SMEs, in strategically selecting and combining smart technologies with lean and green practices for enhanced operational and

sustainability performance. The study is based on a comprehensive literature review and does not involve physical materials or experimental equipment but relies on a structured bibliometric and conceptual methodology. The framework is built by mapping mutual influences among smart technologies (e.g., IoT, automation), lean tools (e.g., Kaizen, value stream mapping), and green practices (e.g., waste reduction, pollution prevention). It categorizes process parameters like waste minimization, energy efficiency, real-time data usage, and continuous improvement, though not in quantitative ranges. Through validation with four SMEs via semi-structured interviews, the framework revealed that combining these paradigms leads to synergies that improve performance metrics such as cost reduction, productivity, and environmental impact. The major technical insight is that Industry 4.0 tools can amplify the benefits of lean and green strategies, transforming them into a cohesive operational ecosystem that promotes sustainability, flexibility, and competitiveness.

Sarkar et al. (2023) have modelled agricultural land suitability for vegetable crop farming using RS and GIS integrated with bivariate techniques in Uttar Dinajpur district, India, with the purpose of improving spatial land utilization for sustainable farming. The study utilized 22 influential parameters including soil properties (pH, nitrogen, OCC), climatic and hydrological features (rainfall, aquifer type, groundwater level), and proximity to infrastructure (Krishi Bazaar, roads), assessed through GIS layers and bivariate models: frequency ratio, entropy index, and weights of evidence. No physical equipment was used, as the work relied on spatial modelling using GIS tools and remote sensing data. The land was classified into four suitability categories: highly suitable (16.93%), moderately suitable (26.64%), marginally suitable (20.38%), and very less suitable (9.34%). ROC curve validations yielded strong predictive power: 0.927 (frequency ratio), 0.888 (entropy), and 0.907 (weights of evidence). The study established that soil characteristics and infrastructure access are primary determinants, providing a strategic planning base for policymakers to enhance crop productivity and regional food security through informed land management.

Li (2025) has explored green wireless networks for future 6G systems, aiming to design an energy-efficient and intelligent communication infrastructure aligned with carbon neutrality goals. The study proposes an architecture for next-generation networks that integrates multi-tiered cloud computing and endogenous intelligence to reduce energy consumption while enhancing network service capabilities. Materials discussed include AI models like federated learning, intelligent protocol stack configurations, and semantic transmission strategies; while physical materials are not used, the virtual architecture is key. Equipment in focus includes network base stations and virtualized computing environments within 6G RANs. Process parameters involved in the energy-efficient design include energy-saving protocols, AI-driven load balancing, and semantic-aware multi-rate data transmission, evaluated across hierarchical network layers. The study models a hierarchical deployment of AI models using federated learning to reduce redundant processing and energy demands, demonstrating that placing AI tasks closer to the data source minimizes overall power usage. Simulation-based results confirmed that endogenous intelligence significantly lowers base station energy use, which currently accounts for 73% of mobile network energy consumption. A novel semantic signaling framework also contributed to reduced transmission redundancy and improved spectral efficiency. These innovations reinforce the transformative potential of 6G architectures in achieving carbon neutrality by embedding AI at the protocol level and distributing computation through federated cloud systems.

Raja et al. (2025) have characterized natural fibers extracted from the *Lawsonia inermis* plant stem for sustainable development and composite material applications, focusing on structural, mechanical, and antibacterial properties to evaluate their industrial viability. The study aimed to explore *Lawsonia inermis* fibers as eco-friendly, biodegradable alternatives to synthetic reinforcement materials. Materials included raw stems of *Lawsonia inermis*, from which fibers were extracted and subjected to antibacterial and structural analyses. Equipment used involved X-ray diffraction (XRD) for crystallinity evaluation, Fourier transform infrared spectroscopy (FTIR) for chemical composition, scanning electron microscopy (SEM) for surface morphology, and mechanical testing instruments to measure tensile strength. Process parameters assessed included fiber concentration (50 μg and 100 μg) in antibacterial tests and analysis under controlled environmental conditions. The results revealed significant antibacterial activity against *E. coli* with inhibition zones of 18 mm and 20 mm, a crystallinity index of 45% suggesting a semi-crystalline structure, and a tensile strength of 19.5 MPa, indicating sufficient strength for moderate load-bearing composite applications. SEM analysis showed grooved surface topography favorable for polymer matrix bonding, while FTIR confirmed the presence of lignin, cellulose, and hemicellulose. This research demonstrated that *Lawsonia inermis* fibers possess a compelling combination of mechanical resilience, antibacterial efficacy, and surface bonding capabilities, making them a viable and sustainable choice for reinforcing biodegradable packaging, structural composites, and biomedical materials.

Al Amin et al. (2025) have examined the synergistic integration of Industry 5.0 technologies with green supply chain management (GSCM) to promote sustainable development in Bangladesh's Ready-Made Garment (RMG) industry. The objective of this research is to identify key factors that facilitate the convergence of advanced industrial

technologies—such as artificial intelligence, IoT, collaborative robots, and big data—with green supply chain practices to achieve sustainability goals. No physical materials or equipment were directly tested, as the methodology centers on a structural modeling approach. Process parameters are defined through interpretive structural modeling (ISM), where nine critical synergistic factors are evaluated and categorized across seven hierarchical levels using level partitioning and MICMAC analysis. These include digital traceability, energy efficiency, circular production models, and predictive analytics for emissions control. The study applies expert input from the RMG sector to validate the framework, highlighting how Industry 5.0 acts as a catalyst for monitoring, implementing, and optimizing environmental performance. The most significant technical insight lies in the creation of a prioritization model to systematically implement digital strategies for environmental compliance and operational efficiency. This strategic alignment enables the RMG sector to adopt a transformative approach to sustainability while addressing global decarbonization mandates and compliance with international green standards.

Setti et al. (2025) have developed a sustainable non-enzymatic electrochemical glucose sensor fabricated from cork-derived laser-induced graphene (cLIG) for urine analysis, aiming to provide an eco-friendly, cost-effective alternative for diabetes monitoring. The study focused on transforming natural cork sheets into functional sensors through direct laser writing (DLW), which simultaneously enabled graphitization and the formation of copper oxide (CuO) nanoparticles by pre-soaking the cork in aqueous $CuSO_4$ solution. Materials included natural cork, $CuSO_4$, and a CO_2 laser for fabrication, forming a porous graphene matrix (Cu-cLIG) embedded with electrocatalytically active CuO NPs. No additional solvents, templates, or post-processing were used, and the process operated under ambient conditions with minimal waste. Process parameters involved current–voltage electrochemical characterizations using cyclic voltammetry (CV) and chronoamperometry (CA) in phosphate-buffered saline and artificial urine, under varying glucose concentrations. The fabricated sensor demonstrated a sensitivity of 223 μA mM^{-1} cm^{-2} and a detection limit (LOD) of 9.7 μM, along with high selectivity against common interfering species and stability over multiple cycles. The fully laser-written sensor offered reproducible and stable performance in artificial urine, highlighting its biocompatibility and utility in non-invasive diagnostics. By eliminating the need for enzymes and fossil-derived substrates, this innovation supports a shift towards scalable green electronics, aligning functionality with environmental sustainability for future health-monitoring technologies.

Waqar et al. (2025) have proposed an integrated management system (IMS) framework to enhance sustainable construction development through coordinated quality, safety, and risk management. The purpose of this study was to overcome the fragmentation in current sustainable construction practices by creating a unified platform capable of capturing and leveraging multi-dimensional data for effective decision-making. The materials involved were quantitative survey responses from 119 construction professionals, while the methodological equipment consisted of structural equation modeling (SEM) and software-based validation techniques like Cronbach's alpha, Fornell-Larcker criterion, and Smart-PLS. Key process parameters analyzed included internal consistency ($\alpha = 0.72$–0.95), path coefficients for quality management ($\beta = 0.643$), risk management ($\beta = 0.53$), and safety management ($\beta = 0.439$), all with $p < 0.001$. Results demonstrated strong positive correlations among the IMS components and Sustainable Construction Management (SCM) dimensions. Digital technologies such as Building Information Modeling (BIM), IoT sensors, data analytics, and digital twins were identified as pivotal in enabling real-time monitoring, scenario simulation, and data-driven interventions. This study established that IMS, when embedded with advanced digital tools and collaborative strategies, can redefine traditional project lifecycles into iterative, optimized workflows that reduce emissions, improve resource efficiency, and enhance stakeholder coordination in the construction sector.

Overall, research across materials science, agriculture, construction, and industrial systems emphasizes sustainable resource use, circular economy principles, and digital integration for efficiency and environmental impact reduction. Studies on zinc extraction from low-grade ores highlight that controlled pre-calcination enhances leaching efficiency, minimizing energy use and unwanted byproducts. In agriculture, smart land-use modelling, digital training tools, and automation improve crop productivity, food security, and sustainability. Life cycle assessments of natural versus artificial materials reveal that optimizing energy sources, transportation, and production methods can significantly lower carbon footprints in manufacturing. Research on green supply chains and smart production frameworks demonstrates that combining lean, green, and Industry 4.0/5.0 technologies—such as AI, IoT, robotics, and predictive analytics and enhances operational performance, reduces emissions, and supports circular production models. Advances in sustainable materials, including natural fibers and cork-derived sensors, showcase eco-friendly alternatives with mechanical, antibacterial, and electrochemical functionality. In wireless networks, integrating AI-driven protocols and semantic communication reduces energy consumption, aligning with carbon neutrality goals. Finally, integrated management systems in construction and industry enable data-driven, real-time decision-making that enhances resource efficiency, safety, and sustainability. Collectively, these studies underscore the importance of controlled processes, digital

technologies, and material innovation in promoting circular resource use, operational efficiency, and environmental stewardship across sectors.

Smart cities and green infrastructure

Santamouris (2014) has reviewed reflective and green roof mitigation technologies to combat urban heat island effects and improve thermal comfort. The purpose was to synthesize theoretical and experimental evidence on how these roof systems reduce urban temperatures, energy demand, and pollution. The study analyzed simulation and field data from cities worldwide, using materials such as single-ply membranes (EPDM, PVC, TPO) and liquid-applied coatings for cool roofs with albedos up to 0.85, and extensive or intensive vegetated systems for green roofs with varying plant species, soil depth, and moisture content. Equipment and tools included mesoscale climate models (MM5, WRF, CSCRC, EnviMet) and experimental temperature measurement setups. Process parameters examined included roof albedo changes (0.1–0.35), vegetation type, latent heat contribution, and building height, with climatic impact assessed as average and peak ambient temperature reduction. Test results indicated that cool roofs can lower average urban ambient temperatures by 0.1–0.33 K per 0.1 albedo increase and peak temperatures by up to 0.9 K, while green roofs on a city scale can reduce temperatures by 0.3–3 K, although benefits are negligible on high-rise buildings. Comparative studies showed that reflective roofs often yield greater peak daytime cooling and higher overall mitigation potential, whereas green roofs provide enhanced night cooling, insulation benefits, stormwater management, and biodiversity. When properly designed and implemented according to urban form and climate, both technologies contribute significantly to heat island mitigation, energy conservation, and improved comfort, with reflective roofs excelling in direct thermal load reduction and green roofs adding multifunctional environmental value.

Albino et al. (2015) have reviewed definitions, dimensions, performance, and initiatives of smart cities to clarify conceptual ambiguity and establish key characteristics, metrics, and best practices for urban smartness. Using an extensive literature review of peer-reviewed studies and official institutional documents, the study examined the evolution of the term "smart city" from its early ICT-focused meaning to a broader socio-technical and governance-oriented framework. Various perspectives—including technology integration, human and social capital investment, governance, and environmental sustainability—were compared, with emphasis on six primary dimensions: smart economy, smart mobility, smart environment, smart people, smart living, and smart governance. The research discussed performance measurement systems such as the Vienna University's six-dimension index, Intelligent Community Forum metrics, and multi-layer models incorporating environmental and innovation layers, as well as the challenges of data availability, indicator weighting, and comparability across cities. Technical aspects included the identification of "hard" domains (infrastructure, energy, transport, waste) where ICT plays a decisive role, versus "soft" domains (education, culture, governance) where ICT supports but does not dominate outcomes. The analysis concluded that a city's smartness depends on integrating technological infrastructure with participatory governance, fostering human capital, ensuring environmental stewardship, and establishing standardized metrics to monitor and guide sustainable, inclusive urban growth.

Kaya (2025) has developed an advanced geo-analytic decision system for optimizing micro-mobility routes to promote sustainable and safe urban transportation infrastructure. The study addresses the growing demand for efficient integration of micro-mobility vehicles such as e-scooters and bicycles into urban environments while enhancing safety and accessibility. Materials include geospatial datasets representing 33 factors such as cyclist lanes, terrain inclination, traffic density, public transport proximity, and recreational zones. The methodology employs Geographic Information Systems (GIS) alongside novel weighting techniques—Fuzzy Logarithm Methodology of Additive Weights (FLMAW) and Fuzzy Simple Weight Calculation (fuzzy SIWEC)—to construct suitability maps. Equipment involved includes GIS platforms and statistical analysis tools like Spearman's correlation and Random Forest classifiers. Key parameters include spatial weight coefficients, route classifications, and route priority indices assessed using the Dombi Bonferroni (DOBI) performance method. Results identified 21 optimized micro-mobility routes, with strong positive correlations ($R > 0.7$) between spatial factors and user preferences. Routes were further categorized into nine usage types, such as commuting or recreational, enabling user-specific planning. The standout technical contribution is the hybrid decision framework, which integrates machine learning and fuzzy logic to create scalable and adaptable micro-mobility infrastructure. This decision system presents a practical roadmap for policymakers to improve urban sustainability and traffic safety simultaneously.

Khan et al. (2025) have developed a BIM-based plugin to optimize sustainable alternatives in value engineering (VE) decision-making for buildings, aiming to improve cost efficiency and reduce embodied carbon through automated material selection within construction projects. The study introduces a plugin integrated into Building Information Modeling (BIM) environments, incorporating lifecycle cost (LCC), energy use, and embodied carbon assessments using APIs linked with Revit software. Materials used included case-based construction datasets embedded within the

plugin to simulate different VE options. Equipment comprised programming platforms such as Visual Studio with C#, Python, and C + + as the development languages, while supporting tools like Dynamo and Revit Macro Manager facilitated advanced customization. Process parameters evaluated included material performance metrics, carbon emission factors, and cost values across multiple design iterations. A case study demonstrated a 10% reduction in project costs and a 15% reduction in embodied carbon compared to conventional approaches. The major technical innovation is the seamless integration of carbon and cost optimization within a single BIM platform, enabling data-driven and real-time sustainable decision-making. This framework enhances traditional VE by introducing automation, streamlining the selection process, and enabling accurate environmental impact assessments that align with evolving green construction standards and regulatory sustainability mandates.

Firoozi et al. (2025) have investigated the strategic integration of solar-heated pavement systems (SHPS) to enhance resilience and sustainability in urban infrastructure, focusing on mitigating winter hazards and promoting energy efficiency. The study explored the design, material selection, and operational efficiency of pavements embedded with heat transfer fluid (HTF) circuits powered by solar energy. Materials used included asphalt and concrete enhanced with graphite and phase change materials (PCMs) to improve thermal conductivity and durability under freeze–thaw conditions. Equipment utilized encompassed computational fluid dynamics (CFD) tools, thermal sensors, and experimental testbeds simulating varying winter climates. Key process parameters studied involved specific heat capacity, thermal conductivity (evaluated using HTFs like glycol–water mixtures), fluid flow dynamics (laminar vs. turbulent), energy storage capability, and radiation efficiency ($\eta = \alpha(1 - \epsilon)$). Results from simulations and field trials showed a 40% reduction in energy consumption and a 60% decrease in winter maintenance costs, with enhanced safety and longer pavement lifespan. Pavement-integrated photovoltaic thermal (PIPVT) modules achieved an energy efficiency of 37.31%, optimizing both heat capture and electrical output. The major technical breakthrough is the dual-functionality of SHPS in both snow-melting and urban heat island mitigation, supported by passive solar energy and optimized piping geometry. This research establishes SHPS as a scalable and environmentally beneficial alternative to conventional winter road maintenance, supporting smart city development and long-term urban sustainability.

Qi et al. (2025) have investigated the effects of green façade retrofitting on thermal performance and energy efficiency of existing buildings in northern China by conducting comparative experiments on four identical laboratory rooms located in Binzhou, Shandong Province. The study aimed to evaluate the impact of different green façade systems on reducing energy consumption and improving indoor thermal comfort. The materials included climbing plants—*Parthenocissus quinquefolia* and *Humulus scandens*, with installation on movable metal grid shelves to simulate direct and indirect façade configurations. Key equipment comprised solar radiometers, Agilent data collectors, automatic thermohygrographs, and 48 type K thermocouples. The process parameters were controlled under two conditions: cooling and non-cooling, with air conditioners set to 24 °C based on ASHRAE standards during cooling trials. Experiments were conducted in July and August 2024, recording data such as wall surface temperature, indoor and outdoor temperature, relative humidity, and solar radiation. The indirect green façade with *Parthenocissus quinquefolia* achieved the highest surface temperature drop of 23.1 °C and reduced indoor temperatures by 1–5 °C across test conditions. Energy savings peaked at 45.75% in this configuration, while direct green façades showed significantly lower performance, as low as 6.43%. It was found that indirect façades outperformed direct ones in terms of both insulation and energy efficiency. These findings underscore the significant technical advantage of indirect green façades in retrofitting scenarios, particularly in hot summer conditions, suggesting their broader applicability in sustainable urban development strategies across climates similar to Shandong's.

Soni et al. (2025) have explored the application of nature-based solutions (NBS) through the sponge city (SC) concept to address urban flooding and climate resilience in Karachi. This research aimed to assess how NBS strategies, such as integrating permeable surfaces, green infrastructure, and stormwater management, could be adapted to megacities in South Asia, with Karachi as the case study. The study reviewed successful SC implementations in China and Saudi Arabia, utilizing case comparisons and spatial data analysis. Key tools and materials used included Shuttle Radar Topography Mission (SRTM) Digital Elevation Models, Landsat imagery from 1990 to 2024, and ArcMap 10.8 for Land Use Land Cover (LULC) assessments. The research did not involve laboratory equipment but applied geospatial analysis methods. Critical SC elements such as rain gardens, porous pavements, retention lakes, green roofs, and walls were evaluated, along with their hydrological benefits and design adaptability. The study identified key technical and policy challenges, including retrofitting existing dense urban areas, policy integration, maintenance demands, and the need for stakeholder coordination. Despite these challenges, examples from Shanghai, Shenzhen, and Riyadh demonstrated tangible benefits including improved flood mitigation, urban cooling, biodiversity, and water retention. Karachi was found to face similar urbanization and climatic stresses as these cities, indicating a high potential for SC-based retrofitting to enhance urban resilience, reduce

vulnerability to climate extremes, and transform its urban water management practices.

Overall, research on smart cities and green infrastructure emphasizes integrating technology, nature-based solutions, and sustainable design to improve urban resilience, energy efficiency, and liveability. Strategies such as reflective and green roofs, green facades, and solar-heated pavements help mitigate urban heat islands, enhance thermal comfort, reduce energy demand, and provide multifunctional environmental benefits including stormwater management and biodiversity. Advanced decision-making tools, such as GIS-based micro-mobility optimization systems and BIM-integrated sustainable material selection plugins, enable data-driven planning for safer, energy-efficient, and low-carbon urban infrastructure. Nature-based approaches like sponge city concepts, permeable surfaces, and green infrastructure improve flood resilience and water management in dense urban areas. Collectively, these studies highlight that combining technological innovation, ecological interventions, and participatory planning can drive sustainable, adaptive, and human-centered urban development.

Urban sustainability

Alami et al. (2023) have reviewed additive manufacturing in the aerospace and automotive industries: recent trends and role in achieving sustainable development goals with the aim of evaluating how 3D printing technologies support sustainability and innovation in these sectors. The study focuses on additive manufacturing (AM) techniques such as selective laser melting, fused deposition modelling, and stereolithography, using materials like polymers (ABS, PLA), ceramics, and metal alloys. Equipment discussed includes 3D printers capable of layer-by-layer fabrication, heat sintering, or chemical bonding systems. The process parameters involve material type, build orientation, energy input, and environmental impact per printed component. While the paper does not present new experimental data, it consolidates technical literature to compare material properties (e.g., heat resistance, strength, recyclability) and application-specific performance metrics. It emphasizes that AM significantly reduces waste and energy consumption, with up to 70% reduction in material usage compared to subtractive methods, and enables lightweight, durable component design critical to both industries. The study highlights that AM supports several Sustainable Development Goals (SDGs), including responsible consumption, industry innovation, and climate action, by minimizing emissions and enabling localized production. By assessing material-process-application fit, it affirms AM's transformative role in achieving cost-effective, agile, and eco-friendly manufacturing solutions.

Chrapek et al. (2023) have developed and examined the sustainable synthesis and manufacturing of silver powder-based metal clay for zero-waste technology applications, with the aim to establish an environmentally friendly and cost-effective method of producing metal clay. The study utilized high-purity metallic silver (99.999%), nitric acid, sodium carbonate, and methylcellulose as primary materials. Equipment employed includes a quartz tube furnace, SEM (JCM-6000Plus), and Scott volumeter. Process parameters comprised silver powder production via solid-state hydrogen reduction, using Arcal gas (95% Ar + 5% H_2) at 200 °C over 3 h, and methylcellulose content in clay ranging from less than 1% to over 3%. The produced silver powder had particle sizes between 0.5 and 4.5 µm. Test results showed that lower methylcellulose content improved the tensile strength (up to 100 MPa), electrical conductivity (up to 80% of bulk silver), and surface roughness (under 1 µm), while higher methylcellulose reduced shrinkage (down to below 5%). A Material Circularity Indicator (MCI) of 0.50 and an E-factor of 0.29 indicated efficient material usage with minimal environmental impact, mainly producing water (84%) and green gases (7%) as byproducts. The study demonstrates that using closed-loop synthesis for silver powders and integrating sustainability metrics leads to cost-effective (€1103.45/kg) and low-waste production suitable for applications like jewellery, electronics, and 3D printing.

Arash et al. (2025) have assessed the environmental impact of 3D printing versus traditional construction techniques in remote housing projects, focusing on the community of Attawapiskat in northern Ontario, Canada. The purpose was to perform a comparative life cycle assessment (LCA) of six construction scenarios involving both conventional building materials and 3D printing technologies using geo-polymer and green concrete. The materials used included traditional concrete, geo-polymer concrete, and locally sourced aggregates, while 3D printing equipment and SimaPro software were employed to model environmental impacts from a cradle-to-gate perspective. Process parameters evaluated comprised material types, transport distances, construction energy, and binder compositions. Results from the LCA showed that using geo-polymer concrete in 3D concrete printing (3DCP) reduced environmental burdens significantly, including up to 85.9% less CO_2 emissions in some applications. Additionally, additive manufacturing allowed for reductions in material wastage and energy use, especially when paired with local resource sourcing. Comparative sensitivity analyses confirmed 3DCP's advantages in reducing greenhouse gas emissions, acidification, smog formation, and fossil fuel depletion. While challenges like printability, machine repair logistics in remote locations, and regulatory acceptance remain, this study highlighted 3DCP's potential to support sustainable housing through reduced carbon footprint, cost savings, and efficient resource use, especially when employing recycled and low-impact materials.

Beraldi et al. (2025) have developed an optimization framework for hybrid multigeneration systems aimed at enhancing sustainability in the residential sector by balancing environmental and economic goals. The study was designed to address the energy challenges of residential buildings by integrating various renewable and non-renewable energy sources, energy storage technologies, and electric vehicle (EV) batteries within a unified multi-criteria decision model. Materials and systems incorporated include photovoltaic panels, solar thermal collectors, biomass systems, combined heat and power units, and lithium-ion storage devices. Equipment utilized was simulation-based optimization software capable of solving bi-objective mixed-integer nonlinear problems involving system configuration, operational scheduling, and life cycle analysis. Key process parameters included CO_2 emission levels, capital and operational costs, energy demand satisfaction, and storage efficiency across hourly and seasonal timeframes. A case study applied to an Italian residential complex showed significant results: up to 65% greenhouse gas (GHG) emission reduction and improved payback periods when compared to traditional energy systems. The major technical contribution lies in the comprehensive system synthesis that includes both embodied and operational emissions through Life Cycle Assessment (LCA), alongside dynamic pricing of energy sources for real-world economic viability. This optimization tool enables strategic planning of residential energy infrastructures aligned with net-zero targets and supports informed investment decisions in hybrid polygeneration systems.

Cotella et al. (2025) have developed a critical success factor (CSF) framework for implementing net-positive housing within the Water, Energy, Food, and Environment (WEFE) Nexus, with a focus on sustainable development in Jordan and South Africa. The study aims to integrate WEFE principles into Sustainable, Innovative, Affordable Housing (SIAH) to enhance resilience and sustainability for low-income communities. Materials include structured questionnaire responses and literature documents, while tools involved unsupervised machine learning algorithms, network analysis using Gephi software, and systematic literature review (SLR) methods. No physical testing apparatus was used as the study was social-science and data-analysis-based. Process parameters analyzed included energy use intensity (EUI), household demand forecasts, housing performance indicators, and net-positive outputs in energy, water, and food. Key findings from clustering and interrelationship mapping of 51 CSFs revealed regional distinctions: in Jordan, priority was given to energy self-sufficiency and affordable material use, while in South Africa, emphasis was placed on food security and water access. Network analysis identified highly interconnected CSFs such as stakeholder engagement, policy integration, technological adaptability, and socio-economic inclusivity. The framework enables policy makers, urban planners, and researchers to systematically plan, evaluate, and implement SIAH-Net Positive strategies by highlighting the key drivers for success across different socio-economic contexts, thereby accelerating the transition to sustainable housing systems in the Global South.

Gergely et al. (2025) have assessed the embodied and operational carbon emissions of residential HVAC systems to provide comprehensive life cycle sustainability benchmarks for climate-conscious building design. The study aimed to quantify and compare the environmental impacts of different HVAC setups including air-to-water heat pumps with both floor heating and radiators, condensing gas boilers with radiators, and pellet boilers with radiators, applied across 20 single-family homes. The material inventory included components of distribution systems and heat producers, while carbon emissions were calculated using Life Cycle Assessment (LCA) principles based on operational data and embodied emissions from manufacturing and installation. Process parameters included embodied carbon values for floor heating (0.464 $kgCO_2eq./m^2/year$) and radiator systems (0.548 $kgCO_2eq./m^2/year$), along with system-level emissions: condensing gas boiler (0.742 $kgCO_2eq./m^2/year$), pellet boiler (1.325 $kgCO_2eq./m^2/year$), and heat pumps with R32 refrigerant (0.972–1.055 $kgCO_2eq./m^2/year$). When substituting R32 with low-GWP R290 refrigerant, emissions dropped to 0.636 and 0.719 $kgCO_2eq./m^2/year$ respectively. The results indicate floor heating offers a lower embodied footprint than radiators, while pellet boilers, despite being renewable-based, have the highest embodied emissions. The research emphasizes that transitioning to heat pumps with low-GWP refrigerants offers the most promising balance between operational and embodied carbon reductions, providing essential insight for policy makers and designers to meet EU carbon neutrality targets in residential buildings.

Pullas et al. (2025) have explored the use of hydrochars derived from exotic fruit residues as sustainable dielectric and insulating materials for electronic and energy applications. The study aims to valorize agricultural waste, specifically residues from mango, mangosteen, moringa, and African oil palm, by transforming them through hydrothermal carbonization (HTC) into green materials suitable for electronics. Materials used include fruit peels and husks subjected to HTC under varying conditions—temperatures of 175–250 °C and retention times of 1–2 h. Equipment includes HTC reactors and dielectric testing setups to measure electrical characteristics. Process parameters such as HTC temperature, residence time, and fruit type are assessed for their influence on relative permittivity (εr) and dielectric loss tangent ($\tan \delta$). Results indicate that moringa husk processed at 175 °C for 1 h yields the most promising dielectric properties, achieving permittivity values of 4.35–6.29 and

extremely low dielectric losses between 0.004 and 0.025. These hydrochars exhibit properties suitable for sustainable substrates in printed circuit boards (PCBs), offering a biodegradable alternative to conventional polymer-insulated materials. This work significantly contributes to green electronics by enabling waste valorization into functional materials, offering both environmental benefits and cost-effective solutions for developing regions.

Superchi et al. (2025) have analyzed the role of green hydrogen as seasonal energy storage in a hybrid renewable energy system for energy autonomy on the island of Tilos, Greece. The study investigates how integrating wind and photovoltaic (PV) energy with battery energy storage systems (BESS) and power-to-gas-to-power (P2G2P) hydrogen technologies can ensure energy independence year-round. Materials include real production and consumption data from an 800 kW wind turbine and a 160 kWp PV system, with additional energy storage technologies involving lithium-ion batteries and hydrogen storage. The simulation was performed using a differential evolution algorithm under varying scenarios, considering three future timelines: current, 2030, and 2050. Key process parameters include storage size scaling, technology mix, and levelized cost of required energy (LCORE), measured in €/MWh. The study reveals that a hybrid system combining BESS and P2G2P achieves a lower LCORE (264 €/MWh currently) compared to battery-only systems (320 €/MWh), with future projections further reducing costs to 132.4 €/MWh by 2050. A major technical highlight is the long-term economic viability and system resilience offered by integrating green hydrogen, which serves as a seasonal storage buffer for renewable energy excess. The findings support the feasibility of green hydrogen as a strategic enabler for decarbonizing isolated grids while improving system flexibility and reliability.

Tufa et al. (2025) have reviewed integrated membrane processes for sustainable lithium extraction with a focus on enhancing selectivity, recovery, and scalability from aqueous brine solutions. The study aimed to critically assess and promote innovative membrane-based strategies capable of overcoming the inefficiencies of traditional solar evaporation methods, particularly in high Mg^{2+}/Li^{+} ratio brines. Materials investigated included various lithium-rich brines sourced from salt lakes, geothermal sources, and reverse osmosis waste, with membrane materials such as nanofiltration (NF), membrane distillation (MD), electrodialysis (ED), Donnan dialysis (DD), supported liquid membranes (SLM), and capacitive deionization (CDI) employed at different stages of extraction. No specific laboratory equipment was directly mentioned, as the paper focused on reviewing prior experimental data and simulation frameworks. The key process parameters analyzed involved membrane selectivity, ion exchange efficiency, water recovery ratios, and energy demands across integrated stages including pretreatment, enrichment, and final crystallization. Test results indicated that membrane integration offers superior ion separation, particularly for reducing Mg^{2+}/Li^{+} ratios, with enhanced environmental performance and water conservation compared to evaporative techniques. A techno-economic analysis emphasized the need for optimization in module configurations and operational costs to scale-up these systems. This research demonstrated that stage-wise, hybrid membrane system design combining electro-driven and non-electro-driven processes can substantially improve lithium recovery efficiency and environmental sustainability, supporting the shift toward green mining and circular resource use.

Zhao et al. (2025) have investigated the impact of ESG performance and green technology innovation on corporate value in Chinese industrial firms, using comprehensive panel data from 2012 to 2023 to explore the causal mechanisms and moderating/mediating effects involved. The purpose of the study was to determine how environmental, social, and governance (ESG) performance influences enterprise value and whether green technology innovation acts as a mediator while R&D investment moderates this relationship. Materials used include ESG scores from Hua Zheng Securities and green patent application data from CNRDS, while financial variables such as ROA, ROE, R&D investment, and green patents were employed for modelling. Equipment is not physical in this empirical study but refers to statistical tools including a two-way fixed effects regression model. The process parameters involve variables like firm age, board independence, R&D expenditure (log-transformed), and ESG rating (1–9), analyzed under fixed-effects assumptions with robust standard errors to address heteroskedasticity. The results demonstrated that ESG positively influences firm value (with coefficients of 0.0038 for ROA and 0.0066 for ROE), green technology innovation significantly mediates this effect (correlation of 0.0562), and R&D investment acts as a negative moderator. Notably, only non-state-owned and small/low-growth firms derive significant value from ESG performance. This empirical evidence reinforces the notion that ESG performance promotes innovation and enhances firm valuation while offering practical implications for firms and regulators aiming to balance sustainability with profitability in China's industrial sector.

Overall, research highlights diverse strategies for promoting urban sustainability across manufacturing, construction, energy, housing, and resource management. Additive manufacturing and 3D printing are shown to reduce material waste, energy use, and emissions while enabling lightweight, durable components and localized production. Sustainable material synthesis, such as silver powders and hydrochars from agricultural residues, demonstrates low-waste, cost-effective pathways for electronics, 3D printing, and industrial applications. In construction, 3D concrete printing with geo-polymer and green concrete significantly

lowers carbon emissions and material consumption, particularly when using local resources. Hybrid energy systems integrating renewables, storage, and green hydrogen optimize residential and island energy infrastructures, balancing operational efficiency, cost, and environmental impact. Life cycle assessments of HVAC systems and residential buildings underscore the importance of low-GWP refrigerants, efficient heating methods, and integrated energy planning to reduce embodied and operational emissions. Policy, stakeholder engagement, and socio-economic factors are key to implementing net-positive housing and sustainable, innovative, affordable housing strategies, especially in low-income contexts. Membrane-based lithium extraction and ESG-driven corporate strategies further emphasize circular resource use, green innovation, and financial value creation. Collectively, these studies demonstrate that combining advanced technologies, efficient materials, renewable energy, and systemic planning can drive urban sustainability, lower environmental impact, and support long-term economic and social resilience.

Digital tools in sustainability

Tao et al. (2018) have examined Digital Twin-driven smart manufacturing systems and their enabling technologies. The study aimed to define the architecture, functionalities, and industrial applicability of Digital Twins (DT) for achieving higher efficiency, flexibility, and intelligence in manufacturing. Materials and resources considered include cyber-physical production systems, IoT-enabled sensors, big data platforms, and manufacturing execution systems, integrated through advanced communication frameworks. Equipment encompassed cloud computing servers, industrial robots, and simulation engines capable of real-time interaction with production environments. Process parameters addressed involved synchronization latency (milliseconds to seconds), data update frequency, simulation accuracy (up to 95%), and system adaptability under varying production loads. Results indicated that DT implementation allows real-time monitoring, predictive maintenance, production scheduling optimization, and fault detection, with case studies demonstrating reductions in downtime by up to 30% and production efficiency gains of 20–25%. The major technical insight revealed the necessity of high-fidelity models combined with real-time data analytics to enable self-optimizing production systems capable of autonomous decision-making. By integrating virtual and physical manufacturing elements via bidirectional data flows, the framework proposed supports adaptive, resilient, and resource-efficient smart factories.

Pandiyan et al. (2024) have reviewed the advancements in green IoT for smart grids to pave the path to sustainability. The purpose of this review was to explore recent technological progress in the integration of Green Internet of Things (Green IoT) within smart grid systems to achieve efficient energy management and environmental sustainability. The study did not employ any experimental materials or equipment but systematically analyzed multiple communication and computing technologies including Wireless Sensor Networks (WSN), Cloud Computing, Edge and Fog Computing, and Artificial Intelligence-based analytics, focusing on their roles in smart energy infrastructure. The paper highlighted key process domains like energy-efficient protocols, predictive analytics, and real-time demand response with system parameters ranging from adaptive communication protocols to distributed intelligence architectures. It discussed challenges such as standardization, interoperability, data security, and energy consumption by IoT devices. The study identified edge and fog computing as crucial for enabling real-time monitoring and reducing latency and bandwidth use. Additionally, successful case studies emphasized the viability of integrating Green IoT technologies for smart grid development. The paper offered policy recommendations and outlined emerging trends, asserting that integrating intelligent, green communication frameworks into power systems can significantly optimize energy distribution, enhance system resilience, and support global sustainability goals.

Kayan et al. (2025) have optimized biohydrogen production from agricultural waste using a digital twin framework to support sustainable supply chain management and carbon neutrality. The purpose of this study was to explore the economic and environmental feasibility of transforming agri-residues into hydrogen fuel, using digital modelling and simulation to streamline logistics and maximize impact. Materials included agricultural residues such as rice straw and wheat stubble, with biohydrogen production processes relying on microbial fermentation. Key equipment used for analysis encompassed digital twin simulations, Monte Carlo models, greenfield analysis tools, and network optimization software. The process parameters focused on hydrogen yield efficiency, spatial supply chain optimization, and lifecycle greenhouse gas emissions reduction. Test results highlighted that optimized logistics through digital twin integration could yield annual social savings of approximately USD 34.54 billion, reduce emissions, and lower dependence on fossil-fuel-based hydrogen. The study also emphasized the significance of site selection, residue availability, and policy support for large-scale implementation. With India's net-zero targets and circular economy ambitions as context, this framework demonstrated that digital twins, when applied to biohydrogen supply chains, can unlock system-wide improvements in sustainability, resource efficiency, and industrial decarbonization.

Thomas and Mishra (2025) have formulated an energy-efficient circular smart supply chain model incorporating wastewater treatment and Industry 4.0 technologies for a

sustainable tyre production industry, aiming to reduce environmental impact and maximize profitability. The study proposes an integrated decision-making system using particle swarm optimization (PSO) to determine optimal investment in automation technology (AI), water purification technology (WI), and green technology (GI). Materials and infrastructure used include wastewater treatment units such as Zero Liquid Discharge (ZLD) systems, automation systems with programmable machines, and emissions-reducing technologies powered by renewable energy. Key parameters explored were recyclability index values, technology efficiency factors, emission control rates, water pollution cost coefficients, and varying customer sensitivities to price and sustainability. The model evaluated two operational scenarios: with and without recyclability index-based demand. In the scenario with recyclability considerations, the profit increased by 52.60%, supported by higher demand for recycled products. Sensitivity analysis revealed that investments in AI, WI, and GI significantly lowered production, emission, and pollution costs, while maximizing circularity through 3R strategies (Reduce, Reuse, Recycle). The technical novelty lies in the simultaneous inclusion of customer-driven sustainability preferences and closed-loop logistics in a comprehensive mathematical framework, enabling firms to transition from linear to circular economies. This approach establishes the feasibility of transforming the tyre industry into a low-emission, resource-efficient system while achieving economic viability and environmental compliance.

Yu et al. (2025) have modeled a joint logistics-and-charging-infrastructure network design problem to evaluate the viability of electric trucks as a green transportation solution, aiming to integrate logistics network optimization with charging infrastructure deployment for emissions and cost reduction. The purpose of this study was to assess the economic and environmental feasibility of adopting electric trucks (E-trucks) by developing a mixed-integer linear programming (MILP) model that simultaneously optimizes logistics routing and charging station locations. Materials in the study were data inputs from a real-world logistics network in Nepal, including suppliers, distribution centers, and retailers. While no physical equipment was used, computational modeling and optimization tools formed the analytical backbone. Process parameters included logistics demands, charging station capacities, routing distances, and scenarios involving different levels of electrification and infrastructure investment. The model tested three distinct scenarios with varying levels of E-truck adoption and infrastructure investment. Test results demonstrated that electrification of the logistics system can reduce logistics costs by up to 33.3% and carbon emissions by up to 55.9%. Scenario analyses revealed that strategically locating charging stations in high-traffic zones significantly improves accessibility and cost-effectiveness. The model also underscored the importance of industry-led infrastructure development and highlighted how network design influences E-truck viability. This integrated approach shows that electric trucks, supported by optimized infrastructure, can substantially enhance both economic and environmental performance in green logistics systems.

Overall, studies highlight the transformative role of digital tools in advancing sustainability across manufacturing, energy, and supply chain systems. Digital Twin (DT) technologies enable real-time synchronization between physical and virtual systems, supporting predictive maintenance, production optimization, and lifecycle-integrated product management. High-fidelity models combined with IoT sensors, cloud computing, and simulation platforms enhance operational accuracy, adaptability, and autonomous decision-making, reducing downtime, improving efficiency, and fostering resource-efficient manufacturing. Green IoT and smart grid integration demonstrate significant energy optimization and environmental benefits, with edge and fog computing enabling real-time monitoring, predictive analytics, and reduced latency. In sustainable supply chains, digital twins facilitate optimized logistics, biohydrogen production from agricultural waste, and circular economy strategies, lowering emissions, improving resource efficiency, and generating substantial economic and environmental gains. Additionally, Industry 4.0 technologies applied in circular manufacturing, including AI, automation, and wastewater treatment, enable firms to adopt closed-loop, low-emission operations, while optimized electric vehicle logistics networks show potential for significant cost and carbon reductions. Overall, digital tools collectively support adaptive, resilient, and resource-efficient systems, bridging technological innovation with sustainable industrial and energy practices.

ESG for sustainability

Khan et al. (2023) have examined the relationship between green leadership, environmental knowledge sharing, and sustainable performance in the manufacturing industry using upper echelon theory. The study aimed to assess how green leadership impacts sustainable performance and the mediating role played by environmental knowledge sharing. Utilizing data from 257 respondents across small and medium manufacturing firms in Pakistan, the study employed structural equation modeling with partial least squares (SmartPLS) to analyse the relationships. The study did not involve material samples or experimental equipment but focused on behavioral and structural organizational data, using constructs such as leadership practices, knowledge dissemination, and sustainability indicators. Process parameters

included path coefficients derived from statistical modeling, which demonstrated that green leadership significantly enhances environmental knowledge sharing, and this, in turn, positively affects firms' sustainable performance across economic, environmental, and social dimensions. The mediation analysis confirmed that environmental knowledge sharing significantly channels the effect of green leadership on sustainability. The study concluded that fostering green leadership and promoting an internal culture of environmental information sharing can substantially uplift organizational sustainability, especially in resource-constrained and energy-intensive industries such as those prevalent in South Asia.

Kuku and Arishi (2024) have explored the application of carbon nanotube (CNT)-based advanced materials for sustainable fabrication, water desalination, hydrogen production, and CO_2 reduction. This study aimed to harness the multifunctional potential of CNTs to address global environmental challenges by improving process efficiency and sustainability. The research utilized various synthesis and doping techniques, combining CNTs with materials such as nitrogen, boron, cobalt, and nickel to enhance performance in each domain. No single equipment set dominated; instead, the study referenced a range of chemical, electrochemical, and catalytic apparatus for material functionalization and characterization. In water desalination, CNT-based membranes mimicking biological aquaporins demonstrated high permeability, selective ion rejection, and antifouling capabilities. For hydrogen production, CNTs acted as electrocatalysts in water splitting, where doping with nitrogen and hybridizing with transition metals lowered overpotential and improved catalytic stability. In CO_2 hydrogenation, CNTs served as efficient catalyst supports, enabling selective conversion of CO_2 to fuels like methane and methanol, with further improvements through functionalization and nanoparticle integration. The major technical achievement lies in the versatile use of CNTs for enhancing membrane, electrode, and catalyst functionalities across sustainability-critical processes, confirming their promise as scalable and environmentally beneficial materials for climate and energy applications.

Furtado et al. (2025) have developed green, switchable photochromic molecularly imprinted polymers (PC-MIPs) for enhanced biosensing applications, focusing on synthesizing environmentally friendly, responsive materials for optical sensing with enantioselectivity. The aim of the study was to fabricate PC-MIPs using supercritical carbon dioxide ($scCO_2$) as a green solvent and evaluate their optical and molecular recognition performance for real-time biorecognition. Materials used include (methacryloyloxyethyl)−3′,3′-dimethyl-6-nitrospiro(2H-1-benzopyran-2,2′-indoline) (SPMA) as a photochromic monomer, L-isoleucine (IsoLEU) as the template molecule, and ethylene glycol dimethacrylate (EGDMA) as the crosslinker. Equipment included a UV-light source (365 nm), UV–vis spectrophotometer, and binding capacity test setups to characterize the molecularly imprinted polymers. Process parameters involved two formulations of PC-MIPs with monomer ratios of T:M:C = 1:12:50 and 1:12:100, synthesized under $scCO_2$ at controlled temperatures and pressures. Test results showed that PC-MIP1 (with higher crosslinker content) achieved a maximum binding capacity of 127 mg IsoLEU/g polymer and an imprinting factor of 1.6. Under UV irradiation, the polymers exhibited a reversible colour change from white to bluish-purple, enabling optical detection. Moreover, delayed optical responses were observed upon target incorporation, confirming real-time sensing capabilities. These results confirm the potential of PC-MIPs as sustainable and high-performance materials for biosensors, especially in detecting chiral molecules in aqueous environments.

Pourmadadi Golaki et al. (2025) have examined a green hybrid biomass-integrated compressed air energy storage (CAES) system with PEM hydrogen production under multiple operating modes, aiming to develop a flexible, efficient, and sustainable energy generation platform. The study focused on integrating renewable biomass fuel with CAES, waste heat recovery, and hydrogen generation technologies to meet varying demand profiles while optimizing energy and exergy performance. Materials used include biomass (e.g., wood chips), air (for compression and storage), and water (for PEM electrolysis), with byproducts being electricity, hydrogen, and hot water. Equipment consisted of a biomass gasifier, compressor, CAES tanks, Organic Rankine Cycle units, PEM electrolyzers, and heat exchangers. Process parameters analyzed covered three operational modes—off-peak, mid-peak, and peak—with key variables such as hydrogen production rate (5.65 g/s), hot water flow (7.8 kg/s), and round-trip energy and exergy efficiencies (66.9% and 51.3%, respectively, for CAES). Test results revealed an overall system efficiency of 61%, and the hybrid system performed optimally in off-peak/mid-peak cycles for hydrogen and thermal co-generation. The technical findings emphasized the system's adaptability and environmental performance, particularly its potential to decarbonize power production by effectively storing renewable energy and valorising biomass waste through a clean, integrated loop.

Faruque et al. (2025) have experimented with the adaptive cultivation of halotolerant microalgae *Chlorella sorokiniana* in synthetic oilfield hypersaline wastewater (HSB) integrated with synthetic municipal wastewater (SMW) to assess its feasibility for sustainable biomass generation, nutrient remediation, and bioenergy production. The purpose was to develop an environmentally friendly process for utilizing industrial waste streams for biomass-based energy. Materials included synthetic HSB mimicking actual oilfield brines, SMW replicating tertiary municipal wastewater, and the microalgae strain *C. sorokiniana* (UTEX 1230).

The equipment comprised borosilicate photobioreactors with controlled lighting (1800–1900 LUX), temperature (22 ± 1 °C), and salinity regimes ranging from 0 to 60% HSB. The experimental procedure involved stepwise saline adaptation over successive culture passages. Biomass concentration peaked at 1373 mg/L under 30% HSB loading, with maximum nutrient removal rates of 89.1% for nitrogen and 98.48% for phosphorus. Proximate and ultimate analyses revealed carbohydrate content between 25 and 62.87%, lipids from 3.7 to 30.94%, and protein between 15.37 and 19.81%, while the higher heating value reached 22.84 MJ/kg, surpassing conventional lignocellulosic sources. The most notable technical insight is the enhanced halotolerance and biochemical productivity achieved through gradual salinity acclimatization, affirming the viability of this microalga for integrated wastewater bioremediation and energy generation. By innovatively coupling synthetic HSB and SMW, the study opens a dual pathway for managing hypersaline waste and producing renewable biofuels in a circular economy framework.

Perera and Ranjith (2025) have developed a fully waste-derived green cement binder to replace Ordinary Portland Cement (OPC) for sustainable construction applications. The purpose of this study is to formulate and evaluate a waste-based green cement (WGC) by utilizing industrial by-products such as rice husk ash and mine tailings, without relying on commercial alkaline activators. The binder is activated exclusively using alkaline compounds derived from waste, distinguishing it from traditional geopolymers or alkali-activated cements. Materials include waste-derived aluminosilicate precursors, while testing is conducted using standard cement mix procedures under ambient curing conditions (21 ± 2 °C and 50 ± 10% RH). Process parameters such as molar ratios of SiO_2/Al_2O_3 (3), Na_2O/Al_2O_3 (1), and H_2O/Na_2O (9) are optimized. Mechanical testing shows that WGC achieved 22 MPa after 56 days of curing, while WGC-OPC composites reached up to 42 MPa with a 50% OPC replacement after 28 days. The study also highlights that recast WGC samples retained 6 MPa strength after reuse, proving reusability. Mineralogical and microstructural analyses confirm that waste-based activation yields durable matrices. The innovative feature lies in creating a fully waste-activated binder, providing a scalable pathway to low-carbon construction materials that align with circular economy principles and enable significant reductions in CO_2 emissions.

Jave-Chire et al. (2025) have examined the impact of green marketing mix elements on brand value and sustainability within the footwear industry. The purpose of this research was to assess whether green marketing strategies—specifically green product, green price, green place, and green promotion—contribute to brand value and enhance economic, environmental, and social sustainability. A quantitative and correlational methodology was employed, utilizing structured questionnaires administered to 363 footwear consumers. Data analysis was conducted using partial least squares structural equation modeling (PLS-SEM), which provided robust insights into the relationships among variables. No physical materials or equipment were used, but key process parameters included statistical metrics such as path coefficients to evaluate the significance of green marketing elements. Green product ($\beta = 0.157$), green promotion ($\beta = 0.156$), and especially green place ($\beta = 0.605$) showed significant positive effects on brand value. Furthermore, brand value significantly influenced economic ($\beta = 0.809$), environmental ($\beta = 0.700$), and social sustainability ($\beta = 0.553$). The study demonstrates that integrating green practices into product design, marketing communication, pricing strategies, and distribution channels can elevate brand value while simultaneously achieving broader sustainability objectives. These insights offer a strategic roadmap for companies to align their marketing functions with environmentally conscious consumer expectations and sustainable development goals.

Rathnasabapathy et al. (2025) have developed a composite Space Sustainability Rating (SSR) to promote responsible practices among satellite operators for the long-term sustainability of the orbital environment. The purpose of this work was to introduce a standardized, quantitative metric to incentivize operators towards reducing space debris and improving orbital behavior through a transparent, verifiable mechanism. The materials involved comprised operational datasets and simulation tools, while the process framework relied on a combination of operator-submitted questionnaires, automated modelling, and validation protocols. The SSR is built on six key modules: Adoption of Design and Operations Standards (ADOS), Collision Avoidance Capabilities (COLA), Detectability, Identification and Trackability (DIT), Data Sharing (DS), External Services (ES), and a verification module. No physical equipment was required; rather, software-based scoring tools and compliance databases were utilized. The study applied the rating to multiple mission types and phases, analysing adherence to debris mitigation guidelines, collision probability, data sharing practices, and end-of-life disposal strategies. The major technical contribution is the modular structure of SSR that ensures compatibility across mission profiles while aligning with international sustainability guidelines. This rating mechanism establishes an industry-aligned, voluntary compliance system that offers recognition for sustainable space operations and sets the groundwork for globally coordinated orbital environmental stewardship.

McGregor et al. (2025) have introduced a risk assessment framework for green hydrogen megaprojects, addressing the dual challenge of climate commitment and project viability, with a focus on guiding investment in large-scale hydrogen

infrastructure. The purpose of the study was to integrate megaproject risk dynamics with technology readiness and CO_2 reduction effectiveness to support decision-making in green hydrogen deployment. Materials examined included electrolyser systems for hydrogen production, green ammonia pathways, and large-scale infrastructure plans exceeding €1 billion in capital costs. While not physical, equipment and modelling tools used comprised project simulation, economic forecasting tools, and comparative lifecycle emissions analysis. Process parameters incorporated were project scale (1–5 GW), expected capital costs (€170–240 billion), CO_2 savings per application (e.g., 51 tons CO_2 per ton H_2 in ammonia), and learning rate effects in pioneer versus follow-up projects. Test results demonstrated that green ammonia applications outperformed others in carbon savings, while project risks were amplified in first-wave initiatives due to limited iterative learning. Key technical findings emphasized the need for staged scaling, minimizing technological novelty per project, and context-based application choice to mitigate underperformance and cost overruns. This structured risk-based strategy empowers stakeholders to make informed investments in green hydrogen with balanced environmental and economic benefits.

Overall, research on sustainability highlights a multi-faceted approach integrating organizational, material, energy, and environmental strategies. Studies emphasize the role of green leadership, environmental knowledge sharing, and green marketing in enhancing economic, environmental, and social performance, demonstrating that internal culture and strategic communication significantly influence organizational sustainability. Advanced materials, such as carbon nanotubes and photochromic molecularly imprinted polymers, show promise for water purification, hydrogen production, CO_2 conversion, and biosensing, combining functional efficiency with environmental benefits. Energy systems integrating biomass, compressed air storage, and hydrogen production, alongside adaptive microalgae cultivation for wastewater treatment and bioenergy, illustrate practical solutions for renewable energy generation and waste valorization. Sustainable construction advances through fully waste-derived cement binders offer low-carbon alternatives, while satellite operators and green hydrogen megaprojects benefit from standardized rating and risk assessment frameworks to ensure long-term environmental and economic viability. Collectively, these studies underscore the integration of technological innovation, resource efficiency, and strategic management to achieve holistic, scalable, and sustainable solutions across industries.

Approaches and techniques employed by researchers

In this collection of different research studies, a diverse range of methodologies were used to address various aspects of sustainable technology and practices by the researchers. These methods can be broadly categorized as follows:

Modeling and simulation

Several studies use modeling and simulation. For example, Sarkar et al. (2023) use spatial modeling with GIS and remote sensing. This involves using geographical information systems (GIS) and remote sensing data to analyse and model agricultural land suitability. Different soil, climatic, hydrological, and infrastructural parameters are integrated, without using physical equipment. Shah et al. (2023) use computational fluid dynamics (CFD) simulations. This technique employs numerical analysis to study fluid flow within catalytic converters, optimizing their design for better pollutant reduction. Alam et al. (2023) use Fuzzy Total Interpretive Structural Modelling (TISM) to understand the interrelationships between various factors that drive the implementation of Agriculture 4.0. Prenzel et al. (2025) model a zero-emission utility system, integrating power purchase agreements with thermal energy storage to find cost-effective ways to eliminate greenhouse gas emissions. Valencia-Díaz et al. (2025) formulate an optimization model. This model uses mathematical programming to manage the energy, water, and carbon relationship in microgrids, helping remote communities achieve sustainable development.

Experimental investigation

A number of studies use experiments. Li et al. (2025) conduct experiments on cement mortar, varying the amount of recycled ceramic aggregates to assess how this affects the mortar's microstructure and properties. Osfouri et al. (2025) investigate sustainable structural lightweight concrete, experimenting with different mixes of foam glass aggregates to find an eco-friendly alternative to traditional concrete. Wei et al. (2025) develop a novel metal/biopolymer composite filament for additive manufacturing, testing different binder systems and process parameters.

Review and analysis

Some studies involve comprehensive reviews and analyses. Haleem et al. (2023) review green manufacturing strategies, examining various approaches to minimize environmental impact and enhance resource efficiency. Cotella et al. (2025) develop a critical success factor framework for implementing net-positive housing, identifying the key elements needed for successful sustainable housing projects. Soni et al. (2025) explore the application of nature-based solutions, specifically the sponge city concept, to address urban flooding. Bendig et al. (2023) performed a systematic review on green manufacturing frameworks, categorizing the influences, elements, and results of green manufacturing research.

Life cycle assessment

Gergely et al. (2025) assess the environmental impacts of different HVAC systems using life cycle assessment (LCA). This method evaluates the carbon emissions associated with a product or system throughout its entire life cycle, from manufacturing to operation. Huang et al. (2025) also employ LCA to evaluate hydrogen production.

Optimization frameworks

Beraldi et al. (2025) develop an optimization framework for hybrid multigeneration systems to improve energy efficiency. Firoozi et al. (2025) explore the design, material selection, and operational efficiency of solar-heated pavement systems. Thomas and Mishra. (2025) Formulates an energy-efficient circular smart supply chain model using particle swarm optimization (PSO).

Statistical modeling

Li et al. (2023) employ partial least squares structural equation modeling (PLS-SEM) to understand the factors that influence the adoption of green technology within businesses.

Other methods

Rathnasabapathy et al. (2025) develop a composite spatial sustainability rating; Tufa et al. (2025) review integrated membrane processes; Kayan et al. (2025) optimize biohydrogen production using a digital twin framework; Siahsarani et al. (2025) investigate cooling methods in titanium milling; McGregor et al. (2025) introduce a risk assessment framework for green hydrogen projects; Sommese and Diana (2025) propose a framework for implementing green roofs.

Analytical framework and comparability

To ensure comparability across a heterogeneous literature body, all reviewed studies were mapped to a unified taxonomy following IPCC sectoral groupings—Energy, Industry, Buildings, Transport, Agriculture and Land Use, Waste, and Cross-cutting Enablers such as digitalization, circular economy, and finance. For each technology or intervention, we extracted a standardized set of metrics (Table 1) encompassing:

(A) Climate—net greenhouse-gas (GHG) abatement (tCO_2e yr^{-1}), abatement cost (US\$ tCO_2e^{-1}), and global-warming potential (GWP) per functional unit from reported life-cycle assessments (ISO 14040/44)
(B) Resources—water intensity (L kWh^{-1} or L kg^{-1}), material-circularity indicator (MCI) where reported, and embodied carbon (e.g., kg CO_2e m^{-2} for building materials)
(C) Systems—energy return on investment (EROI) where available, capacity factor (%), storage hours, grid-integration cost indicators, and basic reliability metrics
(D) Maturity—Technology Readiness Level (TRL), Policy Readiness Level (PRL), and Market Readiness (adoption or diffusion stage).

Ranges and interquartile ranges (IQR) are reported for metrics where at least three independent and comparable data points were available. In cases with fewer observations, individual study values or minimum–maximum ranges are presented, with associated uncertainties discussed explicitly. All metrics are standardized to consistent units, and parameters not reported in the source studies are denoted as "NR" (not reported) to transparently indicate information gaps and context-dependent variability.

Enabling policies and financial mechanisms classification and evidence

Enabling policies and financial mechanisms can be classified by instrument type, with evidence strength assessed from studies included in this review.

Table 1 Summary of comparable metrics for technologies

IPCC Sector	Technology	Climate (tCO_2e / FU, or GWP)	Abatement cost (US$/ tCO_2e or economic metric)	Resources (water L/kWh, MCI, embodied C)	Systems (EROI, capacity factor, storage hrs, grid costs)	Maturity (TRL / PRL / Market note)
Energy	PEM water electrolysis (PV-fed)	4.3–6.5 t CO_2e / t H_2 (LCA)	2741–4048/t H_2; BPV > $11 000/t H_2	Water demand noted qualitatively; L/kg NR	Electrolyser load 95–100 %; EROI NR	TRL 6–8 (demo); PRL medium; Market early commercial
Cooling/ Bucoustic Refrigeration (TAR)	Near-zero refrigerant GWP; LCA NR	Cost metrics NR	Working fluid He; water intensity NR	Drive ratio 1.5–6.4 %; ΔT 67–124 °C; COP qualitative	TRL low–medium (lab); PRL low; Market research stage	Al-Mufti et al. (2025)
Buildings/ HVAC	HVAC systator, heat pump)	Embodied C: floor 0.464; radiator 0.548; HP 0.972–1.055; gas 0.742; pellet 1.325 kg CO_2e/ m²/yr	$/t CO_2 NR	MCI NR; embodied C above	Grid integration NR	TRL 9 (mature); PRL medium–high (dep. refrigerant policy)
Energy/ Storage	**Battery Energy Storage System.83 t CO_2 saved per user / yr	0.193 / kWh; LCOH $4.88 / kg H_2	Li availability, recycling concerns; embodied C NR	Battery 6–7 kWh; ~60 % CAPEX share	TRL 9; PRL high; Market mature but supply-chain issues	Sadik-Zada (2023); Roy et al. (2025)
Energy/ Storage	Hydrogen storage (salt caverns, P2G2P)	Abatem/MWh	Water for electrolysis	Cavern mass flow, turnover modelled; grid cost NR	TRL gas storage mature; H_2 demo–early commercial	Hopken et al.; Superchi et al. (2025)
Water& Energy	Geothermal + Desalination + H_2 (RO vs MED)	GWP NR	LCOH/LCOW qualitter recovery rates reported; L/kWh NR	Exergy efficiencies quantified; grid cost NR	TRL RO/MED high; integrated systems pilot demo	Chitgar & Sadrzadeh
Energy/ Industrial	Molten-salt TES + H_2 turbines	GWP NR	OPEX ↓ ~27 % with TES coupling	Embodied C /m² design metric	TRL high (CSP / industrial); PRL med-high	Industrial case examples
Materials/ Construction	Waste-derived aggregates & engineered binders	Cement CO_2 ↓ 622 → 580 kg/t cement (5Cs framework)	lacement 20–30 %; MCI NR	EROI NR	TRL pilot–commercial; PRL regional variance	Perera & Ranjith (2025) ab–pilot; PRL low
Bioenergy/ Waste	Biohydrogen via dark fermentation	GWP NR	Cost NR	2.5–3.2 mol H_2/mol substrate	System-level metrics NR	TRL pilot; PRL low–medium

Price-based instruments

It includes carbon pricing and feebates. The reviewed literature cites a carbon price level of approximately US \$52 tCO_2^{-1} as a benchmark for stimulating innovation and investment in clean technologies. Quantitative assessments of adoption or cost impacts directly attributable to such instruments are limited, representing a key evidence gap.

Quantity-based instruments

This category covers renewable portfolio standards (RPS) or renewable purchase obligations (RPO), as well as mandates such as zero-emission building (ZEB) targets and minimum energy performance standards (MEPS). These measures are frequently identified as important deployment drivers, though comparable quantitative estimates of resulting changes in adoption rates remain scarce.

Standards and codes

Performance-based building codes and extended producer responsibility (EPR) schemes for construction materials are recognized as fundamental enablers of low-carbon and circular material practices. Case studies document improvements in material performance, but causal evidence linking such codes to large-scale adoption is generally lacking.

Financial instruments

Mechanisms such as green bonds, blended finance, concessional credit, and risk-sharing guarantees appear in several studies. Reported outcomes include reductions in operating expenditure of about 27% where power purchase agreements (PPAs) are combined with molten-salt thermal energy storage, and positive net present values for renewable-energy communities (≈€2042–8195 kW^{-1}). Typical reductions in weighted average cost of capital (WACC) are not reported, indicating an area for further research.

Synthesis and research gaps

Existing evidence links a limited number of policy and finance instruments to measurable deployment or cost outcomes, while most lack harmonized quantitative metrics. Priority areas for further investigation include elasticities of

Table 2 Classification of enabling policy and finance instruments and evidence strength

Instrument type	Example instruments (from manuscript)	Evidence strength in manuscript	Observed metric(s)/numbers reported in manuscript	Action taken in revision
Price-based	Carbon pricing, feebates	Limited—benchmark price cited; no causal Δ adoption or Δ LCOE values reported	Benchmark carbon price ≈US \$52 tCO_2^{-1}. No quantitative deployment or abatement data linked directly to these instruments	Instrument type added and described; marked explicitly as a research gap due to limited quantitative evidence
Quantity-based	Renewable portfolio standards/renewable purchase obligations (RPS/RPO); zero-emission building (ZEB) and minimum energy performance standards (MEPS)	Limited—qualitatively identified as deployment drivers; Δ adoption rates not quantified	Examples: India's 175 GW renewable target (100 GW solar, 60 GW wind, 10 GW bio, 5 GW hydro); modelled CO_2 reduction ≈ 30% at > 30% renewable penetration	Category defined; quantitative evidence absent, therefore labelled research gap
Standards & codes	Performance-based building codes; extended producer responsibility (EPR) for materials; cement sector "5 Cs" framework	Limited–moderate—case-specific performance data; no national-scale adoption metrics	Cement case: CO_2 intensity 622 → 580 kg t^{-1} cement with 18% efficiency improvement; material substitution: 20–30% recycled fines or waste sand	Evidence summarized; table entry added; absence of adoption data noted as research gap
Finance instruments	Green bonds, blended finance, concessional credit, risk guarantees, PPAs + TES, renewable-energy communities (RECs)	Moderate for case studies; no typical WACC reductions reported	PPAs + molten-salt TES → OPEX reduction ≈ 27%; TES storage density ≈1300 kWh_{th} m^{-2}; REC NPV ≈ € 2042–8195 kW^{-1} (low–high market)	Quantitative values incorporated; lack of WACC data flagged as research gap

response to carbon pricing, effects of performance standards on adoption rates, and empirical estimation of Δ LCOE, Δ abatement cost, and typical WACC reductions associated with different financial mechanisms (Table 2).

Discussion

The discussion emphasizes technical solutions that advance efficiency, lower emissions, and optimize resource use. Regional evidence shows their scalability and relevance for sustainable growth.

Technical solutions

Renewable energy

- In remote Himalayan regions of India and Nepal, small hydropower demonstrates decentralized electrification potential where transmission losses make grid extension infeasible; hybrid configurations combining induction and synchronous generators with optimized load-governor controls increase system resilience and cost-effectiveness.
- In India and Malaysia, grid-scale renewable integration highlights technical dependencies: India's ~ 1096 GW potential requires advanced storage systems and smart grid frameworks, while Malaysian simulations show that beyond 30% renewable penetration, CO_2 emissions drop by ~ 30%, contingent on grid flexibility and frequency regulation mechanisms.
- Thermoacoustic refrigeration systems, experimentally tuned with drive ratios of 1.5–6.4%, normalized stack positions of 0.175–0.33, and spacing of 3.3 δk, achieve ΔT up to 124 °C; these parameters illustrate the role of precise acoustic–thermal coupling in delivering sustainable cooling for high-demand tropical climates.
- Photovoltaic-based renewable energy communities in European urban settings yield net present values between €2042 and 8195/kW, with equity-driven profit-sharing mechanisms linked to self-consumption ratios and internal trading prices, thereby aligning distributed PV adoption with energy justice and grid-load reduction.

Hydrogen sustainability

- Biohydrogen production via dark fermentation of rice husk and sugarcane bagasse yields up to 2.5–3.2 mol H_2/mol substrate. Pilot facilities in Tamil Nadu (India) and São Paulo (Brazil) show potential to scale into municipal waste-to-energy frameworks, reducing agricultural residue burning.
- Large-scale green hydrogen hubs in Western Australia (Asian Renewable Energy Hub, 26 GW capacity) and Saudi Arabia (NEOM, 4 GW electrolysis) employ risk frameworks that model price volatility, electrolyzer degradation, and carbon policy changes, enabling investors to project 20–25% IRR stability.
- For energy-importing regions like Japan and South Korea, such frameworks secure long-term hydrogen import contracts, helping offset their reliance on LNG and coal while aligning with 2050 net-zero targets.

Carbon-neutrality

- In China, carbon neutrality depends on cutting coal's share from ~ 84 to < 20% by 2060 through large-scale PV, wind, nuclear, CCUS, grid reinforcement, long-duration storage, and hydrogen carriers, supported by carbon pricing (~ US$52/t$CO_2$) to drive investment.
- In Egypt's cement sector, the 5Cs framework enables staged CO_2 reductions (622 → 580 kg/t cement) via 18% efficiency gains, alternative-fuel substitution, and CCS integration, with monitoring through energy management systems.
- In European and Asian chemical parks, coupling PPAs with molten-salt TES (~ 1300 kWhth/m^2) and hydrogen-fired turbines provides continuous steam and power, lowering OPEX by ~ 27% and buffering renewable variability under volatile hydrogen prices.
- Cross-sector tools strengthen feasibility: fuzzy-QFD improves efficiency (+ 20%) and environmental performance (+ 25%), CFD-based catalytic converter optimisation enhances pollutant conversion, and Mat-STAP anticipatory assessment ensures sustainable feedstock choices at early R&D stages.

Sustainable construction

- Controlled substitution of waste-derived fine and coarse aggregates delivers strong performance gains while cutting embodied impacts. Optimal replacement ranges are ~ 20–30% for waste foundry sand and recycled ceramic fines, and up to 20% for recycled crushed glass, which improves compressive and flexural strength and reduces permeability and embodied CO_2 without significant workability loss. Foam-glass coarse aggregate at ~ 20% volume replacement yields structural-grade concrete with bulk density below 2000 kg/m^3 and compressive strength ≥ 17 MPa, enabling lighter, thermally insulating elements.
- Engineered binders and recycled fines enable ductile, lower-carbon composites for structural use. LC^3-based engineered cementitious composites with recycled fine aggregates can increase tensile ductility by around

80% while keeping long-term compressive strength within ~ 20% of control mixes, supporting crack control and durability in retrofit and new-build applications.

- Connection detailing and joint geometry materially affect structural resilience and service behavior. Modified anchorage holes with adhesive bulbs markedly increase load capacity and produce sustained post-peak load plateaus, and hooked dowel detailing in precast beam–column connections greatly improves energy dissipation and delays stiffness degradation under cyclic (seismic) loading, allowing precast systems to achieve monolithic-like performance.
- Implementation requires rigorous materials control, durability management, and context-aware design. Limit replacement rates where alkali–silica reactivity is a risk and apply mitigation for glass-derived aggregates; monitor powder/aggregate grading and contaminant levels; adopt standardized test protocols for porosity, ASR, and long-term transport properties; and prioritize local circular feedstocks and factory-scale preprocessing to ensure consistent quality and maximize environmental and economic benefits in regional contexts.

Green manufacturing

- Additive processes cut raw-material waste (> 80%) and shorten supply chains, benefiting aerospace and precision hubs in the US, Germany, and France, and enabling distributed, on-demand production for remote markets such as Australia and parts of Canada.
- High energy intensity (e.g., SLS ~ 50–60 MJ/kg) requires pairing AM with heat-recovery, optimized scheduling, and low-carbon power; practical deployments favor regions with strong renewables or cheap clean electricity like Germany, Denmark, Spain, Chile, and Australia.
- Robust circularity—multi-cycle powder reuse with controlled blending and advanced hydrometallurgical recycling that recovers > 90% of critical metals—cuts material costs and dependence on imports, with immediate relevance for battery and alloy industries in China, the EU, and Japan.
- Integrated process innovations (digital twins/AI for in-process control, biopolymer binders for greener extrusion, and $scCO_2$ + MQL cooling) raise part quality and tool life, accelerating scalable low-carbon manufacturing in biomedical and aerospace clusters across Europe, North America, and Japan.

Manufacturing innovation

- IoT and big-data driven monitoring in German automotive factories cut scrap rates by 15% and reduced machine downtime by 18%, saving €1.5–2 million annually per plant.
- Additive manufacturing using powder bed fusion with recycled titanium alloys reduced virgin material demand by 45%, successfully implemented in aerospace supply chains in France and the UK.
- Metal/biopolymer composite filaments used in 3D printing of stainless-steel components lowered energy input per part by 20–25%. Brazilian pilot plants report 25% lower costs for medium-scale production compared to traditional milling.
- Sustainable machining with cryogenic CO_2 cooling extended tool life in titanium milling by 2.8 × and improved surface roughness by 15–18%, with direct application in Japanese dental implant production.

Water treatment

- In Pakistan, electrocoagulation using aluminium and iron electrodes achieves > 90% removal of turbidity, COD, BOD, and heavy metals, with efficiency influenced by current density (5–50 mA/cm^2), electrode spacing (1–5 cm), and water chemistry; hybrid EC–oxidation or membrane systems further enhance scalability.
- In China, microalgae-based wastewater treatment provides dual benefits of nutrient removal (> 90% for nitrogen and phosphorus, 60–80% COD reduction) and biomass yields up to 0.8 g/L, suitable for fuels, fertilizers, or feed; integration with anaerobic digestion and bioelectrochemical systems strengthens energy recovery.
- In Nigeria, molecularly imprinted polymers tailored with ligand-specific cavities deliver highly selective dye removal (> 98% for malachite green and bromocresol green), following pseudo-second-order kinetics and Langmuir adsorption behavior, while maintaining > 85% efficiency in real effluents over multiple cycles.
- In India, nanotechnology-enabled treatment using carbon nanotubes, TiO_2, ZnO, and Fe_3O_4 achieves > 90% removal of dyes, metals, pharmaceuticals, and microbial contaminants under pH 3–11 and nanoparticle doses of 0.1–1.0 g/L, with regeneration potential though scale-up and nano-waste challenges persist.
- In India, green-synthesized ZnO/NiO heterojunction nanocomposites demonstrate multifunctionality, achieving 99% photocatalytic dye degradation, 2.09% DSSC power conversion efficiency, and 262.5 F/g capacitance with 77.8% cycle stability, bridging wastewater remediation with renewable energy applications.

Circular resource use

- In China, replacing natural aggregates with recycled concrete reduced construction emissions by 12 Mt CO_2/year, equivalent to ~ 2% of national cement-sector emissions.
- Hybrid hydrometallurgical-pyrometallurgical steel recycling boosted recovery efficiency to 96%, reducing slag waste by 25%, implemented in Japanese electric arc furnace plants.
- Industrial symbiosis in Kalundborg, Denmark, exchanges steam, gypsum, and wastewater among five industries, saving 240,000 tons of CO_2 annually and reducing freshwater use by 3 million m^3/year.

Smart cities and green infrastructure

- Sponge-city designs implemented in Wuhan, China, increased floodwater retention by 70%, protecting urban districts from 100-year storm events while improving groundwater recharge.
- Singapore's smart water grids achieved 9% reduction in non-revenue water losses, saving 20 million liters per day.
- EV fleet smart charging in California's Bay Area reduced peak electricity demand by 8–12%, avoiding the need for new fossil-fuel peaker plants.

Urban sustainability

- Transit-oriented development in Curitiba, Brazil, reduced transport emissions by 25% and increased public transport ridership to 60% of total trips, serving as a global benchmark.
- Heat-resilient cool roof programs in Ahmedabad, India, cut peak summer temperatures by 2–4 °C, protecting vulnerable populations and reducing air-conditioning loads by 20–22%.
- In Nairobi's Mukuru slum upgrading project, energy-efficient housing prototypes cut household energy bills by 30%, improving affordability while enhancing resilience.

Digital tools

4. Regional deployment of digital twins shows clear contrasts. Germany and the UK use them for predictive aerospace maintenance with ~ 30% downtime reduction, China applies them in manufacturing hubs like Shenzhen for real-time production optimization, and the US demonstrates 20–25% throughput improvements in automotive and energy but struggles with fragmented data ecosystems.

5. Lifecycle applications vary by geography: Japan reduces automotive prototyping cycles by ~ 30% through CAD–IoT integration, South Korea applies DTs in shipbuilding with > 90% failure prediction accuracy, and Denmark/Netherlands optimize offshore wind through DT-based blade fatigue monitoring, showing how regional ecosystems shape efficiency.

6. Green IoT priorities reflect context: India's smart grids in Delhi and Bangalore manage variable demand via edge computing, Kenya and South Africa adopt low-power IoT to extend lifetimes, and Germany/France employ fog computing for renewable integration, revealing location-driven design differences.

7. Circular and logistics models adapt regionally: Canada uses DTs for biomass logistics cost savings, India's Pune tyre industry improves profits > 50% with sustainable automation, Spain/Italy integrate DTs in waste-to-energy, while Norway's EV freight logistics powered by hydropower cut emissions > 56%, demonstrating how geography dictates circular flows and low-carbon logistics outcomes.

ESG for sustainability

- In Pakistan and Europe, leadership-driven ESG adoption and green marketing strategies directly improve energy efficiency by 15–20% in industries and strengthen corporate competitiveness while reducing lifecycle emissions.
- Japan's CNT-based nanomaterials enhance solar cell efficiency beyond 18%, while Europe's supercritical CO_2 polymer processing cuts solvent use by 90%; China and Eastern Europe's waste-activated cements reduce clinker demand, lowering CO_2 emissions by up to 40%.
- Integrated renewable projects, such as Canada's biomass–CAES achieving > 60% round-trip efficiency and India's microalgae systems removing 80–90% COD in wastewater while producing bioenergy, demonstrate high-performance circular models.
- Hydrogen megaprojects in Australia (e.g., 10 GW-scale hubs) and the Middle East drive large-scale decarbonization, while Switzerland's Space Sustainability Rating sets global benchmarks for orbital debris mitigation and responsible innovation.

Implications and limitations of sustainable and green technology

Implications

- Deployment of renewables such as solar, wind, and small hydro supports deep decarbonization in regions like India, Malaysia, and China, where grid integration models show CO_2 reductions of up to 30%.
- Decentralized small hydro and hybrid renewable systems improve rural electrification by balancing demand

with localized generation, reducing transmission losses in remote areas.

- Circular economy pathways emerge through technologies like waste-derived green cement, which replaces up to 50% OPC while achieving compressive strengths of 42 MPa, reducing reliance on virgin resources.
- Hydrogen integration in industrial processes offers CO_2 savings of 51 tons per ton of H_2 in ammonia production, highlighting its role in decarbonizing steel, cement, and chemical sectors.
- Renewable expansion fosters economic growth, with projections in India estimating 330,000 new jobs by 2022 and long-term renewable capacity exceeding 1000 GW potential.
- Innovative solutions such as thermoacoustic refrigeration demonstrate cooling potential up to 124 °C without harmful refrigerants, while CNT-based membranes enhance water desalination and CO_2 reduction efficiency.
- Energy security is reinforced by diversifying supply with biomass-integrated compressed air energy storage (CAES), achieving 61% round-trip efficiency and flexible hydrogen generation.
- Renewable energy communities (RECs) provide equity by distributing financial returns—net present values of €2042–8195/kW—with models reducing cost disparity among consumers.
- Multifunctional systems like microalgae cultivation simultaneously remove > 90% nutrients from wastewater while producing biomass with a higher heating value of 22.84 MJ/kg for biofuel use.
- Collectively, these advances align with SDG 7 (affordable clean energy) and SDG 11 (sustainable cities), while supporting global climate neutrality targets by 2050–2060.

Limitations

- High capital costs remain a barrier, as seen in green hydrogen megaprojects exceeding €1 billion, requiring staged scaling and strong policy backing.
- Renewable intermittency challenges grid stability; for instance, Malaysian models show flexibility constraints when renewable penetration surpasses 30%.
- Geographic limitations restrict deployment—geothermal is region-specific, while small hydro depends on river topography and flow availability.
- Critical resource constraints, such as lithium and rare earth elements, limit the scalability of storage and advanced material technologies.
- Integration of renewables with fossil-heavy grids increases risks of curtailment and requires costly storage and grid reinforcement measures.
- Policy gaps persist, with inconsistent carbon pricing (e.g., ~ US$52/t$CO_2$ suggested) and weak enforcement of sustainability frameworks across regions.
- Recycling technologies for solar panels, batteries, and nanomaterials remain underdeveloped, limiting life-cycle sustainability.
- Some green technologies like additive manufacturing remain energy-intensive, with efficiency gains often offset by high process energy demand.
- Nanomaterials in water treatment achieve > 90% contaminant removal but raise unresolved concerns about toxicity, persistence, and nano-waste impacts.
- Socio-economic inequalities risk being reinforced unless equity-based models, such as income-sensitive REC benefit sharing, are widely implemented.

Causality and cross-sector synthesis

Moving beyond a descriptive catalogue, the review identifies recurring patterns that help explain why certain interventions achieve scale while others remain at the pilot stage. Three cross-cutting structural features consistently emerge: (1) the bundling of technology with policy and finance, (2) procurement and performance standards that create demand-pull, and (3) digital measurement, reporting, and verification (MRV) systems that lower transaction costs for finance and operations.

Evidence across reviewed cases illustrates the importance of bundling. Integrated utility and industrial decarbonization projects, for instance, frequently combine technological installation with long-term contracting and storage solutions—such as zero-emission utility concepts that pair power purchase agreements (PPAs) with thermal energy storage—to reduce operational risk and enhance bankability. Such integrated configurations mitigate deployment uncertainty by aligning revenue streams, enabling staged investment, and sustaining operational continuity.

Procurement frameworks and performance-based standards also function as demand-unlocking mechanisms in construction and materials. When structured CO_2 reduction frameworks or low-carbon binder scoring tools are embedded within procurement specifications, manufacturers and suppliers respond by adjusting production to meet assured market demand, thereby accelerating scale-up. This dynamic is reflected in evidence from the cement decarbonization 5-Cs assessment and low-clinker LC^3 engineered binders.

Digital MRV and digital twin technologies further support scalability by lowering verification costs for financiers and buyers while enabling performance-based contracting. Multiple studies demonstrate that digital twins, IoT-based

Table 3 Cross-sector synthesis matrix

Technology/sector	Key barriers (policy/finance/operations)	Evidence-backed remedies and mechanisms
Utility-scale renewables/industrial decarbonization	Long payback; financing risk; grid integration	Bundled contracts + thermal energy storage + PPAs reduce operational and market risk, aligning revenue with capacity expansion (Prenzel et al.)
Cement and low-carbon binders	Demand uncertainty; inconsistent procurement criteria	Performance-based procurement and 5-Cs assessment frameworks signal demand and guide prioritization of mitigation actions (Khalil et al.)
Sustainable construction materials (recycled aggregates, LC^3, engineered binders)	Quality variability; durability (ASR) concerns	Local preprocessing + standardized QC + moderate substitution ratios (20–40%) maintain reliability while lowering embodied carbon
Digital manufacturing/smart industry/digital twins	High MRV cost; limited transparency; data silos	Digital MRV + IoT-based dashboards provide traceable performance data enabling outcomes-based finance and continuous optimization
Hydrogen and storage systems	High CAPEX; infrastructure coordination risk	Staged project development + integrated risk frameworks + policy support for first-movers mitigate uncertainty and build investor confidence
Water and wastewater systems	Energy intensity; fragmented governance; weak cost recovery	Circular-water and energy-recovery schemes (e.g., biogas, heat recovery) coupled with performance-based utility regulation reduce operating costs and attract green finance
Green manufacturing/industrial processes	Capital lock-in; low resource-efficiency incentives	Resource-efficiency audits + green-procurement and disclosure mandates align operational savings with compliance and reputational gains, encouraging reinvestment in cleaner production lines
Battery recycling/circular materials	Collection inefficiencies; safety and traceability issues; volatile secondary-material markets	Extended-producer responsibility + digital tracking (QR/blockchain) + standardized recycling protocols improve material recovery, transparency, and investor trust in circular supply chains

monitoring, and digital supply-chain dashboards provide reliable performance data that underpin outcome-based finance and operational optimization—mechanisms that reduce transaction costs and facilitate wider adoption. Table 3 summarizes the main barriers and the corresponding evidence-backed remedies identified across sectors.

Equity, justice, and co-benefits

Scaling low-carbon technologies necessitates not only technical feasibility but also social acceptance and distributive fairness. Evidence from photovoltaic-based renewable energy communities (RECs) demonstrates that design elements such as income-sensitive profit sharing and internal pricing structures can reduce cost disparities among participants. As indicated in the reviewed studies, critical equity dimensions include tariff design—particularly the extent to which time-of-use and network charges favor higher-income prosumers—alongside access to retrofit finance for informal or low-income housing and the implications for labour transitions. Workforce re-skilling for low-carbon construction, operation, and maintenance represents an essential enabler of just transitions. The REC analyses reviewed herein, reporting a range of net present values (NPV) and benefit-sharing schemes, illustrate that participatory governance and equitable financial models can embed redistributive outcomes within local energy systems.

Co-benefits to health and well-being further strengthen the case for equitable transitions. Multiple studies reviewed highlight air-quality and health improvements linked to reduced combustion and electrification of heating and mobility. These pathways contribute to lower ambient $PM_{2.5}$ concentrations and corresponding reductions in disability-adjusted life years (DALYs) in urban populations. Although this review does not quantify DALY outcomes directly, the evidence indicates that cleaner end-use technologies—such as heat pumps and electrified transport—yield measurable health benefits. Integrated policy frameworks combining tariff reform, targeted subsidies, and training programmes

are therefore essential to ensure that such co-benefits are equitably distributed and do not reinforce existing socio-economic inequalities.

The water–energy–food–ecosystem (WEFE) nexus introduces additional distributional trade-offs relevant to scaling. As discussed in the reviewed literature, bioenergy expansion may compete for land and water with agricultural production, desalination can enhance water security while generating brine and increasing energy intensity, and data-centre cooling can elevate local water demand unless low-impact cooling alternatives are adopted. These examples underscore the importance of integrated nexus assessments in planning and prioritizing low-carbon pathways, ensuring that technology choices advance not only decarbonisation but also resource equity and social resilience.

Operational realities (sector wise)

Operational constraints determine the pace and reliability of low-carbon transitions across sectors. The reviewed studies highlight several cross-cutting barriers and indicative operational metrics that shape scalability and system performance.

Power systems

High shares of variable renewable energy (VRE) introduce technical and institutional challenges related to grid inertia, resource adequacy, and system flexibility. Evidence from regional case studies indicates that grid integration constraints become pronounced beyond approximately 30% VRE penetration, underscoring the need for enhanced flexibility markets, storage deployment, and interconnection upgrades. Interconnection queue delays and lengthy permitting processes further limit deployment speed, while emerging flexibility markets remain underdeveloped in many jurisdictions. These findings point to the necessity of coordinated planning frameworks that couple transmission expansion with demand response and ancillary service mechanisms.

Buildings and materials

Low-carbon construction materials such as supplementary cementitious materials (SCMs) and limestone calcined clay cements (LC^3) face implementation barriers associated with quality control, curing protocols, and building-code compliance. Reviewed studies report optimal substitution ratios and performance ranges that maintain structural integrity while reducing embodied emissions; however, limited contractor expertise and restricted availability of environmental product declarations (EPDs) impede broader market adoption. Addressing these gaps requires updated standards, certification frameworks, and targeted workforce training to ensure consistent performance of low-carbon materials at scale.

Water systems

Operational realities in the water sector include the high energy intensity of advanced treatment and desalination processes, as well as the environmental management of concentrated brine. Hybrid membrane systems reviewed in the literature highlight the trade-off between treatment efficiency and specific energy consumption, typically in the range of 3–5 kWh m^{-3} for seawater desalination. Scalable solutions will require integration of renewable energy sources, improved brine valorisation strategies, and digital monitoring to detect and minimize leakage losses across distribution networks.

Manufacturing and industrial processes

In energy-intensive industries, fuel switching for high-temperature operations is constrained by process redesign requirements and limited infrastructure for green hydrogen transport and storage. Studies of steel and cement production emphasize logistical bottlenecks for hydrogen delivery and the need to balance retrofitting with deeper process transformation or carbon capture integration. Illustrative metrics include storage densities of thermal energy systems (~1300 kWh_th m^{-2}) and operational expenditure reductions of up to 27% achieved through hybrid storage and efficiency upgrades. Collectively, these insights underscore the importance of coordinated supply-chain development and clear standards for low-carbon industrial fuels.

Urban systems

Urban decarbonisation efforts are shaped by data governance, operational capacity, and spatial constraints. The adoption of digital twins and other smart-city platforms improves predictive maintenance and resource efficiency but introduces data privacy and interoperability challenges. Nature-based solutions (NbS) such as green corridors and sponge-city infrastructure offer co-benefits for resilience but require sustained operation and maintenance commitments and careful management of land-use conflicts. Effective governance mechanisms and stable funding models remain critical to ensuring long-term functionality and equitable distribution of benefits.

Technology deployment horizons

Technologies identified in this review can be broadly categorized by their indicative deployment horizons. Fast wins (≤ 2030) include measures that can be scaled rapidly through existing infrastructure and proven policy mechanisms, such as rooftop and utility-scale photovoltaics, energy-efficiency retrofits, circular construction materials using recycled ceramic or foam-glass aggregates, and low-power IoT systems for smart grids and real-time monitoring. No-regrets or medium-term options (to 2040) encompass technologies requiring moderate system change or investment sequencing, including hybrid and microgrid configurations, hydrogen pilot hubs demonstrated in Malaysia, district-scale nature-based infrastructure such as sponge cities, and advanced recycling of batteries and industrial residues. Long-lead or transformational actions (to 2050) involve capital-intensive interventions that depend on new infrastructure and regulatory frameworks—most notably carbon capture, utilization, and storage (CCUS) in cement and steel, full green-hydrogen integration in heavy industry, and large-scale long-duration storage solutions such as salt-cavern hydrogen systems. This temporal framing clarifies near-, mid-, and long-term priorities for coordinated policy and investment planning.

Uncertainty and sensitivity

The techno-economic performance of low-carbon technologies is highly sensitive to underlying financial and market assumptions. The reviewed literature reports substantial variation in key parameters, including the levelized cost of hydrogen (LCOH) and electricity (LCOE), the weighted average cost of capital (WACC), and carbon pricing levels, with several studies citing a representative carbon price around US$50–60 t^{-1} CO_2. Sensitivity analyses across these parameters consistently demonstrate that the relative attractiveness of competing mitigation options shifts under different macroeconomic and policy contexts.

Three broad patterns emerge. First, low renewable energy costs combined with favorable financing conditions (low WACC) strongly enhance the competitiveness of electrification pathways and renewable hydrogen production. Second, elevated carbon prices improve the viability of carbon capture and storage (CCS) and hydrogen-based decarbonisation routes, particularly in hard-to-abate industrial sectors. Third, spikes in commodity and material prices constrain the feasibility of capital- and material-intensive options, such as certain advanced supplementary cementitious materials (SCMs) or mineral-based carbon capture processes.

A structured qualitative sensitivity framework can therefore clarify how technology rankings evolve under plausible future conditions. Illustrative scenario comparisons within the reviewed studies indicate that shifts in financial and policy parameters can re-order mitigation preferences even without technological breakthroughs. Quantitative tornado charts or harmonised cross-technology sensitivity analyses would require access to the original model datasets; nevertheless, the synthesis presented here delineates the primary drivers of uncertainty and highlights the need for transparent reporting of parameter assumptions in future techno-economic assessments.

Conclusion

The reviewed studies collectively demonstrate a decisive global shift toward embedding sustainability as a central pillar of technological and industrial development. Across the energy, construction, manufacturing, water, and urban sectors, the integration of renewable energy, green materials, digital innovation, and circular resource strategies signals that sustainability has evolved from a peripheral objective to a defining driver of competitiveness and innovation. The evidence underscores how cleaner energy systems, resource-efficient production, resilient infrastructure, and intelligent digital solutions are converging to reshape industrial practices and urban lifestyles along low-carbon trajectories. Importantly, the transition is multidimensional—environmental, economic, and social. It generates green employment, improves public health through cleaner air and reduced emissions, and enhances resilience in regions most exposed to climate risks. This interconnected transformation reveals systemic feedback: progress in one domain, such as hydrogen technologies or circular resource use, reinforces advancements in others, accelerating collective movement toward carbon neutrality and sustainable growth. Taken together, these insights position the adoption of sustainable technologies not merely as an environmental necessity but as a strategic lever for long-term economic vitality and societal well-being. Achieving scale will require coherent policy alignment, reform of public and private procurement mechanisms, and deployment of robust digital monitoring, reporting, and verification (MRV) systems. The sectoral intervention matrix developed in this review provides a practical framework to prioritize such high-leverage actions and translate pilot-scale innovation into system-level transformation.

Future scope

The future scope of research and development in sustainable and green technology is vast and promising, as global priorities continue to shift toward ecological preservation, climate resilience, and sustainable economic growth. With

environmental challenges becoming increasingly complex and urgent, several areas stand out for future exploration:

- *Advanced renewable energy systems:* Continued innovation in solar, wind, tidal, and geothermal energy will focus on improving efficiency, storage capacity, and integration with smart grids. Emerging technologies such as perovskite solar cells and offshore wind farms offer exciting areas for further research and commercialization.
- *Sustainable materials and circular economy:* Future studies will increasingly target biodegradable materials, sustainable packaging, and resource-efficient product design. Emphasis will also grow on circular economy models that minimize waste and maximize reuse and recycling, thereby reducing environmental impact.
- *Green construction and smart cities:* Sustainable urban development, including green architecture, low-carbon building materials, and intelligent infrastructure, will become critical as urban populations rise. Research will focus on integrating Internet of Things (IoT), AI, and data analytics to create energy-efficient smart cities.
- *Carbon capture and climate engineering:* Technologies for carbon capture, utilization, and storage (CCUS), along with direct air capture and geoengineering, are gaining interest as methods to combat climate change. Their effectiveness, scalability, and environmental implications remain key areas for future study.
- *Sustainable transportation:* With the global push toward net-zero emissions, future research will center on electric vehicles (EVs), hydrogen fuel technologies, and efficient public transport systems. Innovations in battery chemistry, charging infrastructure, and vehicle-to-grid integration are especially critical.
- *Green finance and policy innovation:* There is growing recognition of the need for financial mechanisms and regulatory frameworks that support sustainable development. Future studies may analyse the effectiveness of green bonds, ESG investment trends, and climate policy impacts on technological adoption.
- *Agricultural and water sustainability:* Research will expand into sustainable farming practices, precision agriculture, and water conservation technologies. These areas are essential for food security and ecosystem health amid changing climate conditions.
- *Artificial intelligence for sustainability:* AI and machine learning can optimize energy consumption, predict environmental impacts, and drive decision-making in sustainability planning. Future studies will explore their role in automating and enhancing green technologies.

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